



BANARAS HINDU UNIVERSITY

B.Sc.(Hons.) Courses

Effective from 2024 under NEP

Offered by

Faculty of Science



BANARAS HINDU UNIVERSITY



Department of Statistics

Faculty of Science

SYLLABI

B.Sc. (Hons.) Programme

In

Statistics

COURSE STRUCTURE FOR 4-YEAR UG STATISTICS, NEP-2020 Department of Statistics, Banaras Hindu University		
Semester-I		
Course Code	Title	Credits
STAMJ11	Descriptive Statistics	3
STAMJ12	Practical based on STAMJ11	1
STAMN11	Descriptive Statistics	3
STAMN12	Practical based on STAMN11	1
STAMD11	Elementary Statistics	2
STAMD12	Practical based on STAMD11	1
STASE11	Data analysis using Excel	1
STASE12	Practical based on STASE11	2
VA11	(Offered by the University/Institute)	2
VA12	(Offered by the University/Institute)	2
AE11	(Offered by the University/Institute)	2
Total		20
Semester-II		
STAMJ21	Introduction to Probability Theory	3
STAMJ22	Practical based on STAMJ21	1
STAMN21	Introduction to Probability Theory	3
STAMN22	Practical based on STAMN21	1
STAMD21	Inferential Statistics	2
STAMD22	Practical based on STAMD21	1
STASE21	Data analysis using SPSS	1
STASE22	Practical based on STASE21	2
VA21	(Offered by the University/Institute)	2
VA22	(Offered by the University/Institute)	2
AE21	(Offered by the University/Institute)	2
Total		20

Semester-III		
STAMJ31	Basics of Probability Distributions	3
STAMJ32	Sampling Distributions	3
STAMJ33	Practical based on STAMJ31, STAMJ32	2
STAMV31	Connecting Data with Probability Models	3
STAMV32	Practical based on STAMN31	1
STAMD31	Basics of Sample Surveys and Design of Experiment	2
STAMD32	Practical based on STAMD31	1
STASE31	Data analysis using R	1
STASE32	Practical based on STASE31	2
AE31	(Offered by the University/Institute)	2
Total		20

Semester-IV		
STAMJ41	Statistical Inference I	3
STAMJ42	Sample Surveys and Design of Experiment	4
STAMJ43	Applied Statistics	4
STAMJ44	Practical based on STAMJ41,	1
STAMJ45	Practical based on STAMJ42 and STAMJ43	2
STAMN41	Statistical Inference I	3
STAMN42	Practical based on STAMN41	1
AE41	(Offered by the University/Institute)	2
Total		20

Semester-V		
Course Code	Title	Credits
STAMJ51	Operations Research	4
STAMJ52	Numerical Methods	3
STAMJ53	Statistical Inference II	4
STAMJ54	Practical based on STAMJ51, STAMJ52 and STAMJ53	3
STAIN51	Internship	2
STAMV51	Structure and Visualization of Data	3
STAMV52	Practical based on STAMV51	1
Total		20

Semester-VI		
STAMJ61	Stochastic Processes	3
STAMJ62	Statistical Process Control and Reliability	3
STAMJ63	Econometrics and Actuarial Statistics	4
STAMJ64	Official and Vital Statistics	3
STAMJ65	Practical based on STAMJ61, STAMJ62, and STAMJ64	3
STAMV61	Data-Driven Decision Making	3
STAMV62	Practical based on STAMV61	1
Total		20

Semester-VII		
STAMJ71	Linear Models & Regression Analysis	3
STAMJ72	Measure & Probability Theory	3
STAMJ73	Survey Sampling	3
STAMJ74	Advanced Statistical Inference	4
STAMJ75	Practical based on STAMJ71	1
STAMJ76	Practical based on STAMJ73 and STAMJ74	2
STAMN71	Linear Models & Regression Analysis	3
STAMN72	Practical based on STAMN71	1
Total		20
Semester-VIII (for Honours Students)		
STAMJ81	Design and Analysis of Experiments	3
STAMJ82	Multivariate Analysis	3
STAMJ83	Bayesian Inference	3
STAMJ84	Statistical Aspects of Machine Learning	3
STAMJ85	Practical based on STAMJ81	1
STAMJ86	Practical based on STAMJ82, STAMJ83, STAMJ84	3
STAMN81	Design and Analysis of Experiments	3
STAMN82	Practical based on STAMN81	1
Total		20

Semester-VIII (for Honours with Research Students)		
<i>Major Elective theory Courses</i>		
STAMJ8R1	Distribution Theory	3
STAMJ8R2	Technical Demography	3
STAMJ8R3	Bayesian Inference	3
STAMJ8R4	Advanced Theory of Sample Surveys	3
STAMJ8R5	Statistical Process and Quality Control	3
<i>Student may opt any ONE course among the above list of major elective courses with 3 credits</i>		
STAMJ8R6	Practical based on Elective STAMJ8R1 to STAMJ8R5	1
STADS8R1	Project/Dissertation	12
STAMN8R1	Same as the Major Elective opted	3
STAMN8R2	Practical based on STAMN8R1	1
Total		20

Note: Practical will based on the different software and programming languages.

Semester I			
Course Title: STAMJ11: Descriptive Statistics			
Category of Course: Major			
Credits &Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives			
1. To understand the concept of basic statistics, types of data, collection methods, variables and their types. 2. To know the summarization and representation of numerical data through diagrams and graphs, frequency distribution. 3. To understand central value and dispersion of data and their different measurements and describing nature of frequency curve. 4. To aware about bivariate data and their relation/functional relation through different measures.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Origin of Statistics & its application area, Data and its sources: primary and secondary. Data collection methods: designing a questionnaire and schedule. Scales of measurements. Types of variable: discrete variable, continuous variable, qualitative variable, quantitative variable. Construction and properties of tables. Distribution of variables. Discrete distribution, continuous distribution, cumulative distribution.		10
II	Graphical summarization of data: histogram, frequency polygon, frequency curve, ogives, stem-and-leaf plot, box plot, bar chart, multiple bar chart, stack bar plot, pie chart, line chart, diagrammatic and graphical representation of grouped data, frequency and cumulative frequency distribution and their applications.		10
III	Concept of central tendency and its measures, partition values, dispersion and relative dispersion, moments, Sheppard’s correction for moments, skewness, kurtosis and their measures. Bivariate data: Scatter diagram, measurement of covariation, standardized covariation (product moment correlation coefficient) with assumptions and its properties, Spearman's correlation, point biserial correlation, biserial correlation, coefficient of determination.		10
IV	Theory of attributes: Independence and Association of attributes, various measures of association for two-way classified data.		5
V	Regression: purpose and scope, simple linear regression and concept of error, principle of least square and fitting of linear regression. Properties of regression equations, properties of regression coefficients, predicted and residual variances.		10
Texts / References:			
1. Chattamvelli, R., & Shanmugam, R. (2023). <i>Descriptive statistics for scientists and engineers</i>. Springer. 2. Holcomb, Z. (2016). <i>Fundamentals of descriptive statistics</i>. Routledge. 3. Kore, B. G., & Dixit, P. G. (2022). <i>Descriptive Statistics-I. Nirali Prakashan, Pune</i>. 4. Goon A.M., Gupta A.K. and Das Gupta B. (2014): <i>Fundamental of Statistics, Vol. I</i> (5th edition), World Press, Calcutta. 5. Mood A.M., Greybill, F.A. and Boes D.C. (1974): <i>Introduction to the Theory of Statistics</i>, McGraw Hill. 6. Freund, R. J., & Wilson, W. J. (2003). <i>Statistical Methods</i>, Elsevier. 7. Croxton F.E., Cowden D.J. and Karlin S. (1973) <i>Applied General Statistics</i>.			
Learning Outcomes: Upon completing this course, students will achieve the following learning outcomes:			
1. Students will learn data types and tools: become familiar with both quantitative and qualitative data, and the statistical tools available for their analysis. 2. They will learn to tabulate and present them into diagrams and graphs and interpret data. 3. Understand Statistical Measures: Comprehend various measures of central tendency, dispersion, skewness, kurtosis, and the properties of moments. 4. Explore Linear Correlation: Determine linear correlations between two variables using different measures, and study their properties. 5. Apply the least squares method to fit curves and regression lines, and understand their basic properties 6. Compute Measures of Association: Learn to calculate different measures of association within the theory of attributes.			

Course Title		STAMJ12: Practical Based on STAMJ11	
Paper will be based on numerical problems on different units of Paper STAMJ11			
Unit			Hours of Practical
Unit I&II			5
Unit III			10
Unit IV&V			15
Course Title:		STAMN11: Descriptive Statistics	
Category of Course: Minor			
Credits &Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives 1. To understand the concept of basic statistics, types of data, collection methods, variables and their types. 2. To know the summarization and representation of numerical data through diagrams and graphs, frequency distribution. 3. To understand central value and dispersion of data and their different measurements and describing nature of frequency curve. 4. To aware about bivariate data and their relation/functional relation through different measures.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Origin of Statistics & its application area, Data and its sources: primary and secondary. Data collection methods: designing a questionnaire and schedule. Scales of measurements. Types of variable: discrete variable, continuous variable, qualitative variable, quantitative variable. Construction and properties of tables. Distribution of variables. Discrete distribution, continuous distribution, cumulative distribution.		10
II	Graphical summarization of data: histogram, frequency polygon, frequency curve, ogives, stem-and-leaf plot, box plot, bar chart, multiple bar chart, stack bar plot, pie chart, line chart, diagrammatic and graphical representation of grouped data, frequency and cumulative frequency distribution and their applications.		10
III	Concept of central tendency and its measures, partition values, dispersion and relative dispersion, moments, Sheppard’s correction for moments, skewness, kurtosis and their measures. Bivariate data: Scatter diagram, measurement of covariation, standardized covariation (product moment correlation coefficient) with assumptions and its properties, Spearman's correlation, point biserial correlation, biserial correlation, coefficient of determination.		10
IV	Theory of attributes: Independence and Association of attributes, various measures of association for two way classified data.		5
V	Regression: purpose and scope, simple linear regression and concept of error, principle of least square and fitting of linear regression. Properties of regression equations, properties of regression coefficients, predicted and residual variances.		10

Texts / References: 1. Chattamvelli, R., & Shanmugam, R. (2023). <i>Descriptive statistics for scientists and engineers</i>. Springer. 2. Holcomb, Z. (2016). <i>Fundamentals of descriptive statistics</i>. Routledge. 3. Kore, B. G., & Dixit, P. G. (2022). Descriptive Statistics-I. <i>Nirali Prakashan, Pune</i>. 4. Goon A.M., Gupta A.K. and Das Gupta B. (2014): Fundamental of Statistics, Vol. I (5th edition), World Press, Calcutta. 5. Mood A.M., Greybill, F.A. and Boes D.C. (1974): Introduction to the Theory of Statistics, McGraw Hill. 6. Freund, R. J., & Wilson, W. J. (2003). Statistical Methods, Elsevier. 7. Croxton F.E., Cowden D.J. and Karlin S. (1973) Applied General Statistics.			
Learning Outcomes: Upon completing this course, students will achieve the following learning outcomes: 1. Students will learn data types and tools: become familiar with both quantitative and qualitative data, and the statistical tools available for their analysis. 2. They will learn to tabulate and present them into diagrams and graphs and interpret data. 3. Understand Statistical Measures: Comprehend various measures of central tendency, dispersion, skewness, kurtosis, and the properties of moments. 4. Explore Linear Correlation: Determine linear correlations between two variables using different measures, and study their properties. 5. Apply the least squares method to fit curves and regression lines, and understand their basic properties 6. Compute Measures of Association: Learn to calculate different measures of association within the theory of attributes.			
Course Title	STAMN12: Practical Based on STAMN11		
Paper will be based on numerical problems on different units of Paper STAMN11			
Unit			Hours of Practical
Unit I&II			5
Unit III			10
Unit IV&V			15
Course Title STAMD11:Elementary Statistics			
Category of Course: MD			
Credits & Hours of Teaching:			
	Theory	Practical	Cumulative
Credits	2	1	3
Hours of Teaching (Total)	30	30	60
Course Objectives: 1. To understand data and its type. 2. To develop idea of graphical representation of data 3. To convert data to single point and its distance 4. To understand bivariate relation			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Data Types and Measurement Scales: Understanding different data types (nominal, ordinal, interval, ratio), Identifying appropriate measurement scales for variables, Implications of measurement scales on descriptive statistics.		5
II	Data Visualization for Descriptive Statistics: Diagrammatic and Graphical representation of data (bar plots, histograms, box plots), Interpreting visual representations of data distributions, using summary statistics in conjunction with graphs. Introduction to Exploratory Data Analysis (EDA): Overview of EDA and its importance in data analysis, The EDA process and its key components, Understanding the role of EDA in decision-making.		5
III	Measures of Central Tendency: Mean median, and mode, weighted mean and its application,		10

	selecting appropriate measures for different data distributions. Measures of Dispersion: Range and interquartile range, Quantiles and percentiles, Variance and standard deviation, Coefficient of variation, Skewness and kurtosis. Summary statistics and data distributions.		
IV	Exploring Relationships and Dependencies: Cross-tabulation and contingency tables, Scatter plots and measures of correlation coefficients (Pearson, Spearman), Interpreting relationships and dependencies between variables. Regression Analysis: Simple and multiple linear regression, interpreting regression coefficients and making predictions	10	
Texts / References 1. Chattamvelli, R., & Shanmugam, R. (2023). <i>Descriptive statistics for scientists and engineers</i>. Springer. 2. Holcomb, Z. (2016). <i>Fundamentals of descriptive statistics</i>. Routledge. 3. Kore, B. G., & Dixit, P. G. (2022). Descriptive Statistics-I. <i>Nirali Prakashan, Pune</i>. 4. Goon A.M., Gupta A.K. and Das Gupta B. (2014): Fundamental of Statistics, Vol. I (5th edition), World Press, Calcutta. 5. Mood A.M., Greybill, F.A. and Boes D.C. (1974): Introduction to the Theory of Statistics, McGraw Hill. 6. Freund, R. J., & Wilson, W. J. (2003). Statistical Methods, Elsevier. 7. Croxton F.E., Cowden D.J. and Karlin S. (1973) Applied General Statistics.			
Learning Outcomes: 1. Student will learn types of variable, nature and scale of measurement. 2. Enhance the understanding data structure and hidden facts in data through diagrams. 3. Student will be capable to know the summary statistics and its utility. 4. Student will be capable to know the idea of relationship between variables and prediction.			
Course Title	STAMD12: Practical Based on STAMD11		
Paper will be based on numerical problems on different units of Paper STADMD11			
Unit		Hours of Practical	
Unit I&II		10	
Unit III		10	
Unit IV		10	
Course Title: STASE11: Data analysis using Excel			
Category of Course: SEC			
Credits &Hours of Teaching			
	Theory	Practical	Cumulative
Credits	1	2	3
Hours of Teaching (Total)	15	60	75
Course Objectives: 1. To understand basics of EXCEL and their functions. 2. To know about data mining, including data importation and organization of data. 3. To understand the statistical analysis using EXCEL.			

Course Content: The course content is divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content	Hours of Teaching	
I	How to use variables; enter data, copy and paste cells, and format text in Excel. Auto sum, fill, sort, filter, find, replace, and conditional formatting functions to your data. Create tables and pivot tables for data handling. Manipulate data with new variables, if statements, and other operators. Use freeze and name functions for data analysis.	5	
II	Create frequency distributions and graphs for your data. Decorate and export your graphs to other applications. Compute descriptive statistics, correlation, regression, and probability with Excel functions. Interpret the results for decision making.	5	
III	Test hypotheses with t tests, chi-square tests, ANOVA, and other methods. Rank and calculate percentiles for your data. Import and export data from different sources to Excel. Link and consolidate data across workbooks and worksheets. Analyze data from multiple sources.	5	
Texts / References 1. Bill Jelen , and Michael Alexander (2021): Microsoft Excel 2019 Pivot Table Data Crunching (Business Skills), Microsoft Publication. 2. Wayne L. Winston (2014): Marketing Analytics: Data-Driven Techniques with Microsoft Excel 1st Edition, Wiley Publication. 3. Gordon S. Linoff(2015): Data Analysis Using SQL and Excel 2nd Edition, Wiley Publication.			
Learning Outcomes: Students will learn to: 1. EXCEL tasks such as data entry, formatting, and creating formulas. 2. Essential data analysis techniques including sorting, filtering, pivot tables, different charts and graphs. 3. Organization of data for applying different statistical tests using EXCEL.			
Course Title	STASE12: Practical Based on STASE11		
Paper will be based on numerical problems on different units of Paper STASE11			
Unit			Hours of Practical
Unit I			20
Unit II			20
Unit III			20
Note: Syllabus of Course code STAVA11, STAVA12 and STAAE11 will be framed at faculty level			
Semester-II			
Course Title: STAMJ21: Introduction to Probability Theory			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives: 1. To understand fundamental knowledge of probability theory and its computation. 2. To explore characteristics of random variables and its probability distribution. 3. To understand expectation, moments and different generating functions. 4. Additionally, it provides insight into key theoretical distributions and their basic properties			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content	Hours of Teaching	
I	Review of combinatorial analysis. Random experiment: Trial, sample point, sample space, elementary events, events, equally likely, disjoint events, mutually exclusive events and exhaustive events. Classical and relative frequency approach to probability, axiomatic approach to probability, merits and demerits of these approaches, properties of probability. Conditional probability and independence, total and compound probability theorems, partition of sample	15	

	space, Bayes theorem and its applications	
II	Random Variable: Concept of discrete random variable, distribution function, probability mass function, joint probability mass function of several discrete random variables, marginal and conditional probability mass functions. Continuous random variable, probability density function, joint density function of two continuous variables, marginal and conditional probability density functions.	10
III	Expectation of random variables and its properties, conditional expectation, moments in terms of expectation, conditional expectations.	5
IV	Moments and cumulants, moment generating function, cumulant generating function, probability generating functions, Probability inequalities (Markov's and Chebychev's).	10
V	Basic idea and properties of some theoretical probability distributions: Binomial, Poisson and Normal distributions.	5
Texts / References - Ross, S. M. (2010). A first course in probability (Vol. 2). New York: Macmillan. 8th edition - Feller, W. (1991). An introduction to probability theory and its applications, Volume 2. John Wiley & Sons. - Bhat B.R., Srivenkataramana T. and Rao Madhava K.S. (1996): Statistics, A Beginner's Text, Vol. I and II, New Age International (P) Ltd. - Snedecor's G.W. and Cochran W.G. (1989): Statistical Methods, Iowa State University Press. 8th edition - Mood A.M., Greybill F.A. and Bose D.C. (2017): Introduction to the Theory of Statistics, McGraw Hill. 3rd edition - Spiegel M.R. (1992): Theory and problem of Statistics, Schaum's Publishing Series. 2nd edition - Goon A.M., Gupta M.K. and Das Gupta B. (2013): Fundamental of Statistics, Vol. I, World Press, Calcutta - Meyer P.L. (1970): Introductory Probability and Statistical Applications, Addison Wesley. 2nd edition - Parzen E. (1992): Modern Probability Theory and its Applications, Wiley Eastern. - Hogg R.V. and Craig A.T. (2019): Introduction to mathematical Statistics, Amerind Publishing Co. 8th edition - Hoel P.G. (1984): Introduction to Mathematical Statistics, Asia Publication. 5th edition - Rohtagi, V.K. (2015) An Introduction to Probability Theory and Mathematical Statistics, John Wiley and Sons. 3rd edition.		
Learning Outcomes: Students will be able to: - Acquire a strong understanding of elementary probability theory and its applications. - Comprehend the axiomatic foundation of modern probability theory and recognize the role of random variables in analyzing random phenomena. - Understand the laws of probability and effectively use Bayes' theorem - Characterize probability models and functions of random variables for both single and multiple random variables. - Evaluate and apply expectation, moments and characteristic functions, and grasp concepts related to inequalities and probabilistic limits. - Gain a thorough understanding of some standard univariate distributions and some properties.		

Course Title	STAMJ22: Practical Based on STAMJ21		
Paper will be based on numerical problems on different units of Paper STAMJ21			
Unit			Hours of Practical
I&II			10
III			6
IV&V			14
Course Title: STAMN21: Introduction to Probability Theory			
Category of Course: Minor			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives:			
1. To understand fundamental knowledge of probability theory and its computation.			
2. To explore characteristics of random variables and its probability distribution.			
3. To understand expectation, moments and different generating functions.			
4. Additionally, it provides insight into key theoretical distributions and their basic properties			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Review of combinatorial analysis. Random experiment: Trial, sample point, sample space, elementary events, events, equally likely, disjoint events, mutually exclusive events and exhaustive events. Classical and relative frequency approach to probability, axiomatic approach to probability, merits and demerits of these approaches, properties of probability. Conditional probability and independence, total and compound probability theorems, partition of sample space, Bayes theorem and its applications		15
II	Random Variable: Concept of discrete random variable, distribution function, probability mass function, joint probability mass function of several discrete random variables, marginal and conditional probability mass functions. Continuous random variable, probability density function, joint density function of two continuous variables, marginal and conditional probability density functions.		10
III	Expectation of random variables and its properties, conditional expectation, moments in terms of expectation, conditional expectations.		5
IV	Moments and cumulants, moment generating function, cumulant generating function, probability generating functions, Probability inequalities (Markov's and Chebychev's).		10
V	Basic idea and properties of some theoretical probability distributions: Binomial, Poisson and Normal distributions.		5
Texts / References			
1. Ross, S. M. (2010). A first course in probability (Vol. 2). New York: Macmillan.8th edition			
2. Feller, W. (1991). An introduction to probability theory and its applications, Volume 2. John Wiley & Sons.			
3. Bhat B.R., Srivenkataramana T. and Rao Madhava K.S. (1996): Statistics, A Beginner's Text, Vol. I and II, New Age International (P) Ltd.			
4. Snedecor's G.W. and Cochran W.G. (1989): Statistical Methods, Iowa State University Press. 8th edition			
5. Mood A.M., Greybill F.A. and Bose D.C. (2017): Introduction to the Theory of Statistics, McGraw Hill. 3rd edition			
6. Spiegel M.R. (1992): Theory and problem of Statistics, Schaum's Publishing Series.2nd edition			
7. Goon A.M., Gupta M.K. and Das Gupta B. (2013): Fundamental of Statistics, Vol. I, World Press, Calcutta			
8. Meyer P.L. (1970): Introductory Probability and Statistical Applications, Addison Wesley. 2nd edition			
9. Parzen E. (1992): Modern Probability Theory and its Applications, Wiley Eastern.			
10. Hogg R.V. and Craig A.T. (2019): Introduction to mathematical Statistics, Amerind Publishing Co. 8th edition			
11. Hoel P.G. (1984): Introduction to Mathematical Statistics, Asia Publication. 5th edition			
12. Rohtagi, V.K. (2015) An Introduction to Probability Theory and Mathematical Statistics, John Wiley and Sons. 3rd edition.			

Learning Outcomes: Students will be able to:			
(1) Acquire a strong understanding of elementary probability theory and its applications.			
(2) Comprehend the axiomatic foundation of modern probability theory and recognize the role of random variables in analyzing random phenomena.			
(3) Understand the laws of probability and effectively use Bayes' theorem			
(4) Characterize probability models and functions of random variables for both single and multiple random variables.			
(5) Evaluate and apply expectation, moments and characteristic functions, and grasp concepts related to inequalities and probabilistic limits.			
(6) Gain a thorough understanding of some standard univariate distributions and some properties.			
Course Title	STAMN22: Practical Based on STAMN21		
Paper will be based on numerical problems on different units of Paper STAMN21			
Unit			Hours of Practical
I&II			10
III			6
IV&V			14
Course Title: STAMD21: Inferential Statistics			
Category of Course: MD			
Credits & Hours of Teaching:			
	Theory	Practical	Cumulative
Credits	2	1	3
Hours of Teaching (Total)	30	30	60
Course Objectives:			
1. To understand the measurement of uncertainty.			
2. To understand the estimation process:			
(i) Point estimation			
(ii) Interval estimation			
(iii) Hypothesis testing			
3. To understand non parametric hypothesis tests			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Overview of statistical methods and their applications. Probability and Probability Distributions: Fundamentals of probability theory, Discrete and continuous probability distributions, Binomial, Poisson and Normal distribution and its properties, Sampling distributions standard error and Central Limit Theorem.		10
II	Statistical Inference-Estimation; Point estimation and interval estimation, Confidence intervals for population parameters, Sample size determination for estimation.		5
III	Hypothesis Testing: Introduction to hypothesis testing, Z-test, One-sample and two-sample t-tests, Categorical Data Analysis: Chi-square test of independence of attributes, Chi-square test of goodness-of-fit, Chi-square test for contingency tables, Log-linear models for categorical data analysis, Interpreting and reporting results for categorical data.		10
IV	Nonparametric Methods: Wilcoxon signed-rank test, Mann-Whitney U test, Kruskal-Wallis test, Interpreting nonparametric test results.		5
Texts / References			
1. Cheryl Ann Willard, Statistical Methods, (2020): An Introduction to Basic Statistical Concepts and Analysis, Taylor and Francis Group, 2020.			
2. Goon A.M., Gupta A.K. and Das Gupta B. (1999): Fundamental of Statistics, Vol. I, World Press, Calcutta.			
3. Mood A.M., Greybill, F.A. and Bose D.C. (1974): Introduction to the Theory of Statistics, McGraw Hill.			
4. Goon A.M., Gupta A.K. and Das Gupta B. (1999): Fundamental of Statistics, Vol. II, World Press, Calcutta.			
Learning Outcomes: Students will learn to:			
1. Estimate unknown parameters of different models on the basis of data and its fitting.			
2. Validate the known parameter value on the basis of model and data.			

3. To use different statistical tests for validation/comparison of quantitative and qualitative data for known distribution.			
4. To use different statistical tests for validation/comparison of quantitative and qualitative data for unknown distribution.			
STAMD22: Practical Based on STAMD21			
Paper will be based on numerical problems on different units of Paper STAMD21			
Unit			Hours of Practical
I			4
II&III			20
IV			6
Course Title: STASE21: Data analysis using SPSS			
Category of Course: SEC			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	1	2	3
Hours of Teaching (Total)	15	60	75
Course Objectives:			
1. To understand basics of SPSS and their different functions.			
2. To know about data mining, including data importation and organization of data through SPSS.			
3. To understand the statistical analysis using SPSS.			
4. To strengthen students' ability to analyze and interpret data using a range of statistical techniques			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	How to use SPSS interface, terms, and data formats? Import and export data from different sources. Navigate data editor and variable view. Data types and measurement levels. Data Entry and Manipulation, Recode and transform variables. Merge and append datasets. Descriptive Statistics: central tendency, variability, etc. Frequency distributions and histograms. Cross-tabulate variables and make contingency tables. Function for descriptive analysis.		5
II	Data Visualization: Create and customize charts and graphs. Add data labels and annotations. Export charts and graphs from SPSS. Transform variables (standardize, normalize, etc.). Create new variables in SPSS. Compute and recode variables. Run frequency analysis and descriptive statistics on specific variables.		5
III	Parametric Tests: Independent and paired t-tests, one-way ANOVA, etc. Nonparametric Tests: chi-square test, Mann-Whitney U test, Wilcoxon signed-rank test, Kruskal-Wallis test, etc. Correlation and Regression: Pearson's and Spearman's correlation coefficients, simple and multiple linear regression analysis. Check assumptions and interpret results of regression analysis. Generate scatter plots and regression output.		5
Texts / References			
1. Kenneth Stehlik-Barry, Anthony J. Babinec (2017): Data Analysis with IBM SPSS Statistics, Packt Publishing.			
2. Andy Field (2020): Discovering Statistics Using IBM SPSS Statistics, Sage Publication Ltd.			
3. Julie Pallant (2013): SPSS Survival Manual: A Step by Step Guide to Data Analysis Using IBM SPSS 5th Edition.			
4. George, D., & Mallery, P. (2019). <i>IBM SPSS statistics 26 step by step: A simple guide and reference</i> . Routledge.			
5. Leech, N. L., Barrett, K. C., & Morgan, G. A. (2014). <i>IBM SPSS for intermediate statistics: Use and interpretation</i> . Routledge.			
6. Cleophas, T. J., &Zwinderman, A. H. (2016). <i>SPSS for Starters and 2nd Levelers</i> (pp. 35-40). Cham, Switzerland: Springer.			

Learning Outcomes: Students will learn to:			
1. SPSS tasks such as data entry, formatting, and creating formulas.			
2. Essential data analysis techniques including sorting, recoding and computing, filtering, different charts and graphs.			
3. Organization of data for applying different parametric and non parametric statistical tests using SPSS.			
Course Title	STASE22: Practical Based on STASE21		
Paper will be based on numerical problems of Paper STADSE21			
Unit			Hours of Practical
I			30
II			20
III			10
Note: Syllabus of Course code STAVA21, STAVA22 and STAAE21 will be framed at faculty level			
Semester III			
Course Title: STAMJ31: Basics of Probability Distributions			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives:			
1. This course is designed to introduce various discrete and continuous probability distributions along with their properties and real-life applications.			
2. This course will also introduce the theoretical development of the distribution of function of one/two-dimensional random variable.			
3. The students will also have idea about bivariate data and associated numerical measures.			
4. To understand the idea and application of bivariate probability distribution.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Standard discrete and continuous distributions: binomial, Poisson, geometric, negative binomial, hypergeometric, uniform, normal, beta, exponential, and gamma distributions.		15
II	Moment generating function, probability generating function of above distributions, bivariate normal distributions with properties and their applications. Additive property and applications of above distributions		10
III	Distribution of function of univariate and bivariate random variables using moment generating function, distribution function and linear transformation method.		5
IV	Correlation and Regression of bivariate continuous random variables. Regression functions and their properties		10
V	Bivariate normal distribution: Marginal distribution, conditional distribution and simple problems based on bivariate normal distribution.		5
Texts / References			
1. Mukhopadhyay, P. (1996): Mathematical Statistics, New Central Book Agency.			
2. Hoel P.G. (1971): Introduction to Mathematical Statistics, Asia Publishing House.			
3. Goon A.M., Gupta A.K. and Das Gupta B. (2014): Fundamental of Statistics, Vol. I (5 th edition), World Press, Calcutta.			
4. Mood A.M., Greybill, F.A. and Boes D.C. (1974): Introduction to the Theory of Statistics, McGraw Hill.			
5. Forbes, C., Evans, M., Hastings, N., & Peacock, B. (2011). <i>Statistical distributions</i> (1st ed.). John Wiley & Sons.			
6. Hogg, R. V., McKean, J. W., & Craig, A. T. (2021). <i>Introduction to mathematical statistics</i> (8th ed.). Pearson Education India.			
7. Feller, W., & Morse, P. M. (1958). <i>An introduction to probability theory and its applications</i> .			
8. Rohatgi, V.K. and Saleh, A.K.M.E. (2001): An Introduction to Probability and Statistics, Second Edition, John Wiley & Sons.			

Learning Outcomes: Students will be able to:			
1. Express the features of discrete and continuous random variables.			
2. know about the different characterizations of discrete probability distribution			
3. know about the different characterizations of the continuous probability distribution			
4. define different generating function for probability distributions			
5. describe bivariate data numerically and graphically.			
6. have some idea about bivariate probability distribution and its applications.			
Course Title: STAMJ32: Sampling Distributions			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives:			
1. To understand idea of population and sample			
2. To understand concept of sampling distribution			
3. To derive sampling distributions of function of random sample			
4. To understand order statistics.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Concept of population and their types, simple random sample from a distribution, statistic and its sampling distribution, standard error, Construction of sampling distribution of statistic; derivation of distribution of sample mean, sample variance and sample proportions		15
II	Derivation of some more sampling distributions viz. Chi-square distribution, t-distribution, F-distribution, Z-distribution and their properties with applications.		15
III	Order statistics: Definition and distributions of different order statistics viz. smallest, largest, r^{th} etc. Joint distribution of order statistics and distribution of Range and median along with their applications.		15
Texts / References			
1. B Mood A.M., Greybill F.A. and Bose D.C. (1974): Introduction to the Theory of Statistics, McGraw Hill.			
2. Freund J.E. (2001) : Mathematical Statistics, Prentice hall of India.			
3. Hoge J.L. and Lehmann E.L. (1964): Basic Concepts of Probability and Statistics, Holden Day.			
4. Snedecor G.W. and Cochran W.G. (1989): Statistical Methods, 8 th edition, Iowa State University Press.			
5. Mukhopadhyay P. (1996): Mathematical Statistics, NCBA.			
6. Wackerly, D. D., Mendenhall III, W., & Scheaffer, R. L. (2014). <i>Mathematical Statistics with Applications</i> (8th ed.). Duxbury Press.			
Learning Outcomes: Upon completing this course, students will achieve the following learning outcomes:			
1. Concept of types of population and methods of drawing the random sample.			
2. Construction of sampling distributions of sample-based statistic viz., mean and proportions.			
3. Properties and applications of ordered random variables.			
4. Idea about distribution of order statistics and function of order statistics.			
5. Theory and applications of some standard sampling distributions.			

Course Title		STAMJ33: Practical Based on STAMJ31 & STAMJ32	
Paper will be based on numerical problems of Paper STAMJ31 & STAMJ32 as per following distribution			
Paper and Unit			Number of Practical
Paper STAMJ31Unit I to V			30
Paper STAMJ32Unit I to III			30
Course Title: STAMV31: Connecting Data with Probability Models			
Category of Course: MV			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives			
1. To make a solid understanding of applications of discrete and continuous probability distributions.			
2. To understand the concept of sampling distributions and the Central Limit Theorems for applications.			
3. Apply probability models to classification and prediction tasks.			
4. Explore advanced probability models and their applications.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Statistical approach of probability. Conditional probability and its application based on data.Idea of probability distributions viz Bernoulli, Binomial, Poisson, Geometric, negative binomial normal, uniform and other continuous distributions and their properties.		15
II	Fitting of distributions on random data sets. Basic idea of inference. Population, Sample, parameter, estimator, estimate, methods of estimation. Probability distribution of functions of sample data. Concept of sampling distribution.		10
III	Idea of central limit theorem and introduction to hypothesis testing and construction of confidence intervals. Confusion matrix and types of errors in hypothesis testing. Level of significance, critical region.		10
IV	Idea of small and large sample tests. Data based application of Z test, t test, paired t test, chi square test, Idea of transition probability Advanced		10
Texts / References			
1. David Spiegelhalter (2020). Art of Statistics, Learning from data. A Pelican Book.			
2. Debra A. Perkowski, Michael Perkowski, UMO University of Missouri, (2006). Data and Probability Connections. Pearson Publication.			
3. Wackerly, D. D., Mendenhall III, W., & Scheaffer, R. L. (2007). <i>Mathematical Statistics with Applications</i> (7th ed.). Duxbury Press.			
4. Hogg, R. V., McKean, J. W., & Craig, A. T. (2012). <i>Introduction to Mathematical Statistics</i> (7th ed.). Pearson Education.			
Learning Outcomes: Students will be able to:			
1. Applications and utility of Probability models and their real-life applications.			
2. Statistical concepts and sampling distributions of statistic.			
3. How to apply probability models for decision making in real life problems.			
4. Learn some advanced probability models and their applications.			
Course Title		STAMV32: Practical Based on STAMV31	
Paper will be based on numerical problems of different units of Paper STAMV31			
Unit			Hours of Practical
I			4
II			10
III			6
IV			10

Course Title: STAMD31: Basics of Sample Surveys and Design of Experiment			
Category of Course: MD			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	2	1	3
Hours of Teaching (Total)	30	30	60
Course Objectives: 1. To explain the concepts of population, sample, census, sampling and non- sampling error 2. To understand concept of probability and non probability sampling 3. To plan, design and conduct experiments. 4. To understand the concepts of ANOVA and basic designs			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Concept of population and sample, need for sampling, complete enumeration versus sampling. Basic concepts in sampling, sampling and Non-sampling errors.		5
II	Concept of probability and non-probability sampling and their applications: Simple random sampling with and without replacement, Stratified random sampling, proportional and optimum allocation, Systematic sampling (when population size (N) is an integer multiple of sampling size (n)).		10
III	Analysis of Variance (ANOVA); One-way ANOVA and post hoc tests, Two-way ANOVA, Assumptions and limitations of ANOVA, Interpreting ANOVA results and performing follow-up analyses.		5
IV	Experimental Design: Principles of experimental design, Completely Randomized Design (CRD), and Randomized Block Design (RBD). Randomized controlled trials, Factorial designs and interaction effects, interpreting experimental design results.		10
Texts / References 1. Cochran, W. G. (1977). <i>Sampling Techniques</i> (3rd ed.). John Wiley and Sons, New York. 2. Sukhatme, P. V., Sukhatme, B. V., Sukhatme, S., & Asok, C. (1998). <i>Sampling Theory of Surveys with Applications</i> (4th ed.). Indian Society of Agricultural Statistics, New Delhi. 3. Goon, A. M., Gupta, M. K., & Das Gupta, B. (2014). <i>Fundamentals of Statistics</i> , Vol. II (5th ed.). World Press, Calcutta. 4. Lohr, S. L. (2021). <i>Sampling: Design and Analysis</i> (3rd ed.). Brooks/Cole, Cengage Learning. 5. Lehtonen, R., &Veijanen, A. (2012). <i>Survey Sampling: Theory and Methods</i> . Springer. 6. Dean, A., & Voss, D. (2023). <i>Design and Analysis of Experiments</i> (7th ed.). Springer.			
Learning Outcomes: Students will learn to: 1. Distinguish between probability and non-probability sampling. 2. Understand the process of selecting sampling units under different sampling schemes. 3. To know about basic principles of design and to conduct the experiments under basic designs. 4. They learn about complex designs like factorial designs and its applications.			

Course Title		STAMD32: Practical Based on STAMD31	
Paper will be based on numerical problems of Paper STAMD31			
Unit			Hours of Practical
I			6
II			6
III			8
IV			10
Course Title: STASE31: Data analysis using R			
Category of Course: SEC			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	1	2	3
Hours of Teaching (Total)	15	60	75
Course Objectives 1. To understand the overview of R and R-studio and their applications. 2. To know about data manipulation, data import and export selecting columns and filtering rows, sorting data, and summarizing data with basic descriptive statistics. 3. To understand Data visualization, basic plots. 4. To write and use R functions for statistical analysis.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Introduction to R: Overview of R and its uses in data analysis, Installing R and R Studio, Introduction to R packages. Basic R syntax and data structures: Using R as a calculator: arithmetic operations, vectors and matrices, data frames for tabular data, indexing and subsetting data structures. Data manipulation: selecting columns and filtering rows, sorting data, adding and modifying variables, summarizing data with basic descriptive statistics.		5
II	Data import and export: reading data from csv and excel files, exporting data to different formats, handling missing values during data import. Data visualization: Basic plots (scatter plots, bar plots, line plots), customizing plot aesthetics (labels, titles, colors), adding multiple plots on one canvas. Control structures and functions: conditional statements (if-else), loops (for, while).		5
III	Writing and using functions: Applying functions to data, basic statistical analysis, descriptive statistics (mean, median, standard deviation), hypothesis tests (t-tests, chi-square tests), correlation analysis, basic regression models.		5
Texts / References 1. John Chambers, Software for Data Analysis: Programming with R, Springer, 2008. 2. Phil Spector, Data Manipulation with R, Springer, 2008. 3. Hadley Wickham, Advanced R, Chapman & Hall/CRC The R Series, 2014 4. W.H. Press, S.A. Teukolsky, W.T. Vellering and B.P. Flannery, Numerical Recipes in C, Second edition, Cambridge University Press, 1993.			
Learning Outcomes: Upon completing this course, students will achieve the following learning outcomes: 1. Install and proficiently use R programming language in R and R Studio and their Mathematical operations. 2. Data visualization, Import, export, data manipulation, functions. 3. Define, calculate, and apply probability and probability distributions to solve diverse problems. 4. Conduct and interpret various hypothesis tests to facilitate decision-making processes. 5. Analyze, interpret, and understand data and their analysis using R.			

Course Title		STASE32: Practical Based on STASE31	
Paper will be based on numerical problems of Paper STASE31			
Unit			Hours of Practical
I			20
II			20
III			20
Note: Syllabus of Course code STAAE31 will be framed at faculty level			
Semester IV			
Course Title: STAMJ41: Statistical Inference-I			
Category of Course: Major			
Credits& Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives			
1. The objective of this course is to introduce the idea about parametric inferential procedures. 2. To introduce the concept of point and interval estimation of the parameter 3. To understand the concept of testing of a hypothesis and its applications in real scenario. 4. To introduce the concept and theoretical development of small & large sample parametric test. 5. To introduce the concept and theoretical development of non- parametric test.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Elementary idea of point estimation of parameters, properties of good estimator: unbiasedness, consistency, efficiency, & sufficiency with simple illustrative examples. Idea of Confidence interval estimation		10
II	Concepts of statistical hypotheses: null and alternative hypothesis, critical region, two types of errors, level of significance, p-value and power of a test. One and two tailed tests. Statement of Law of Large Numbers and Central Limit Theorem.		10
III	Small Sample tests: t-test for single Mean, difference of Means and Paired t-test. chi-square test for Goodness of fit and independence of attributes. F-test for Equality of variances.		10
IV	Large sample Tests: Large sample tests for single mean and difference of two means. Confidence intervals for mean(s). Large sample test for single proportion, difference of proportions, Standard deviation(s) and Correlation coefficient(s).		10
V	Non-parametric tests- Definition, advantages and disadvantages, Sign, Run, Median tests, Spearman's rank correlation test, Wilcoxon Mann-Whitney test, Kolmogorov Smirnov - one sample and two sample tests.		5
Text book/ References:			
1. Mood A.M., Greybill F.A. and Boes D.C. (1974) : Introduction to the Theory of Statistics, McGraw Hill. 2. Freund J.E. (2001) : Mathematical Statistics, Prentice hall of India. 3. Goon A.M., Gupta M.K. and Das Gupta B. (2014) : Fundamental of Statistics, Vol. I (5 th edition), World press, Calcutta. 4. Rohatgi, V.K. and Saleh, A.K.M.E. (2001): An Introduction to Probability and Statistics, Second Edition, John Wiley & Sons. 5. Rao C.R. (1973): Linear Statistical Inference and its Applications, Revised edition, Weley Eastern. 6. Hoges J.L. and Lehmann E.L. (1964): Basic Concepts of Probability and Statistics, Holden Day. 7. Snedecor G.W. and Cochran W.G. (1989) : Statistical Methods (8 th edition), Iowa State University Press. 8. Sheskin, D. J. (2020). <i>Handbook of parametric and nonparametric statistical procedures</i> . Chapman and hall/CRC, Fifth Edition 9. Roussas, G. G. (2015). <i>An introduction to probability and statistical inference</i> . Elsevier, Second Edition 10. Almudevar, A. (2021). <i>Theory of statistical inference</i> . Chapman and Hall/CRC, First Edition 11. Wasserman, L. (2013). All of statistics: A concise course in statistical inference, Second Edition			

Learning Outcomes: After completing this course, the students will be able to understand the followings;			
1. About the point and interval estimation of the parameter(s)			
2. Various properties of a good estimators and their theoretical development.			
3. Concept of large sample, law for large number and related theorem.			
4. Concept of small and large sample parametric test.			
5. Concept and application of various non-parametric test.			
Course Title: STAMJ42: Sample Surveys and Design of Experiment			
Category of Course: Major			
Credits& Hours of Teaching			
	Theory	Practical	Cumulative
Credits	4	1	5
Hours of Teaching (Total)	60	30	90
Course Objectives:			
This course is designed			
1. To provide the concept of sampling and related terminologies.			
2. To introduce the concept of various probability and non-probability sampling techniques and their comparisons.			
3. To introduce the concept of one way and two-way ANOVA.			
4. To describe the need and applications of design of experiments and related principles.			
5. To understand the layout and analysis of various types of designs viz. CRD, RBD and LSD.			
6. To understand the layout and analysis of factorial designs.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Concept of population and sample, need for sampling, complete enumeration versus sampling, Basic concepts in sampling, sampling and non-sampling errors. Acquaintance with the working (questionnaires, sampling design, methods followed in field investigation, principal findings, etc.) of NSSO and other agencies undertaking sample surveys.		10
II	Concept of probability and non-probability sampling. Simple random sampling with and without replacement, estimation of population mean, population proportions and their standard errors. Stratified random sampling, proportional and optimum allocation, comparison with simple random sampling for fixed sample size.		15
III	Systematic sampling (when population size (N) is an integer multiple of sampling size (n)). Estimation of population mean and standard error of this estimate, comparison with simple random sampling.		10
IV	Elementary idea of cluster sampling. Use of auxiliary information, ratio, product and regression methods of estimation.		10
V	Analysis of variance for one way and two-way classifications (with one observation per cell), need for design of experiments, basic principles of experimental design: randomization, replication and local control, complete analysis and layout of completely randomized design, randomized block design and Latin square design. Factorial experiments: 2 ² and 2 ³ factorial experiments.		15
Texts / References:			
1. Sampath S. (2005): Sampling Theory and Methods, Narosa Publishing House, New Delhi.			
2. Cochran W.G. (1977): Sampling Techniques, John Wiley and Sons, New York.			
3. Sukhatme P.V., Sukhatme B.V., Sukhatme S. and Asok C. (1998): Sampling Theory of Surveys with Applications (4th ed.), Indian Society of Agricultural Statistics, New Delhi.			
4. Goon A.M., Gupta M.K. and Das Gupta B. (2014): Fundamentals of Statistics, Vol. II, world Press, Calcutta.			
5. Sampath S. (2000): Sampling Theory and Methods, Narosa Publishing House, New Delhi.			
6. Des Raj (2000): Sample Survey Theory, Narosa Publishing House, New Delhi.			
7. Murthy M.N. (1967): Sampling Theory and Methods, Statistical Publishing Society, Calcutta.			
8. Kish L (1965): Survey Sampling, John Wiley and Sons, New York.			
9. Hansen M.H., Hurwitz W.N. and Madow W.G. (1975) : Sample Survey Method and Theory, Vol. I, Methods and Applications, Vol. II, New York and London, Wiley Publication.			
10. Cochran W.G. and Cox G.M. (1957) : Experimental Design, John Wiley and Sons.			
11. Das, M. N. and Giri J. (1986) : Design and Analysis of Experiments, Springer Verlag.			
12. Kempthorne O. (1965) : The Design and Analysis of Experiments, Wiley Eastern			
13. Singh, R., & Mangat, N. S. (2013). Elements of survey sampling (Vol. 15).			

Learning Outcomes: After completing this course, the students will understand the followings;			
1. Concept of population and different methods of selecting random sample			
2. Concept and idea about probability and non-probability sampling techniques along with their realistic applications.			
3. About different types of methods of estimation through the use of auxiliary information.			
4. Concept and meaning of one way and two-way ANOVA.			
5. The importance of design of experiment and associated principle			
6. Analysis and layout of different types of designs.			
Course Title: STAMJ43: Applied Statistics			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	4	1	5
Hours of Teaching (Total)	60	30	90
Course Objectives: This course is design to			
1. Understand time series analysis and its components: stationary/non-stationary, moving averages, trends, seasonality, cycles, and randomness.			
2. Learn classical decomposition and differencing approach for time series models.			
3. Introduction to time series models like AR, MA, ARMA, and ARIMA.			
4. Study index numbers, price/quantity relatives, composition, and different methods (Laspeyres, Paasche, Marshall-Edgeworth, Fisher, chain base).			
5. Understand consumer demand theory, the law of demand, price elasticity, demand curve estimation, and forms of demand functions.			
6. Analyse income distributions, Pareto and lognormal distributions, Lorenz curve, and Gini coefficient.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Time Series Analysis: A realization of stochastic process, nature of time series data, Stationary and non-stationary time series, moving averages, different components: secular trend, seasonal, cyclic and random		15
II	Classical decomposition model, Differencing approach, time series models: Introduction to some time series models such as AR, MA and ARMA, ARIMA model		8
III	Index Numbers: Price relatives and quantity or volume relatives, Link and chain relatives' composition of index numbers; Laspeyre's, Paasche's, Marshal Edgeworth's and Fisher's index numbers; chain base index number, tests for index number, cost of living index number.		15
IV	Theory and analysis of consumer's demand, law of demand, price elasticity of demand, estimation of demand curves, forms of demand functions, Engel's curve, income elasticity of demand		15
V	Analysis of income and allied distributions: Pareto distribution, graphical test, fitting of Pareto law, illustration, lognormal distribution and properties, Lorenz curve, Gini's coefficient		7
Texts / References:			
1. Brockwell, P. J., & Davis, R. A. (2009). <i>Time Series: Theory and Methods</i> . Germany: Springer New York.			
2. Brockwell, P. J., & Davis, R. A. (2013). <i>Introduction to Time Series and Forecasting</i> . Germany: Springer.			
3. Chatfield, C. (2003). <i>The Analysis of Time Series: An Introduction</i> (6th ed.). Chapman and Hall/CRC.			
4. Mukhopadhyay, P. (1994). <i>Applied Statistics</i> . New Central Book Agency Pvt. Ltd., Calcutta.			
5. Goon, A. M., Gupta, M. K., & Das Gupta, B. (2008). <i>Fundamentals of Statistics</i> , Vol. II. World Press, Calcutta.			
6. Mankiw, N. G. (2021). <i>Principles of Economics</i> (9th ed.). Cengage Learning.			
7. Sen, A. (1973). <i>On Economic Inequality</i> . Oxford University Press.			
8. Cowell, F. A. (2000). <i>Measuring Inequality</i> . Oxford University Press			
Learning Outcomes: After completing this course student will able to :			
1. Analyze and model time series data.			
2. Apply classical decomposition and differencing approaches on time series data.			
3. Utilize the concept of AR, MA, ARMA, and ARIMA models.			
4. Construct and interpret various index numbers.			
5. Understand the consumer demand theory and estimation of demand functions.			
6. Capable to analyze and fit income distributions using Pareto and lognormal distributions, and calculate Lorenz			

curve and Gini coefficient.			
Course Title		STAMJ44: Practical Based on STAMJ41	
Paper will be based on numerical problems of Paper STAMJ41			
Paper and Unit			Number of Practical
I			2
II			4
III, IV & V			24
Course Title		STAMJ45: Practical Based on STAMJ42 &STAMJ43	
Paper will be based on numerical problems of Paper STAMJ42 &STAMJ43			
Paper STAMJ42 Unit I to V			30
Paper STAMJ43 Unit I to V			30
Course Title: STAMN41: Statistical Inference-I			
Category of Course: Minor			
Credits& Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives			
1. The objective of this course is to introduce the idea about parametric inferential procedures.			
2. To introduce the concept of point and interval estimation of the parameter			
3. To understand the concept of testing of a hypothesis and its applications in real scenario.			
4. To introduce the concept and theoretical development of small & large sample parametric test.			
5. To introduce the concept and theoretical development of non- parametric test.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Elementary idea of point estimation of parameters, properties of good estimator: unbiasedness, consistency, efficiency, & sufficiency with simple illustrative examples. Idea of Confidence interval estimation		10
II	Concepts of statistical hypotheses: null and alternative hypothesis, critical region, two types of errors, level of significance, p-value and power of a test. One and two tailed tests. Statement of Law of Large Numbers and Central Limit Theorem.		10
III	Small Sample tests: t-test for single Mean, difference of Means and Paired t-test. chi-square test for Goodness of fit and independence of attributes. F-test for Equality of variances.		10
IV	Large sample Tests: Large sample tests for single mean and difference of two means. Confidence intervals for mean(s). Large sample test for single proportion, difference of proportions, Standard deviation(s) and Correlation coefficient(s).		10
V	Non-parametric tests- Definition, advantages and disadvantages, Sign, Run, Median tests, Spearman's rank correlation test, Wilcoxon Mann-Whitney test, Kolmogorov Smirnov - one sample and two sample tests.		5
Text book/ References:			
1. Mood A.M., Greybill F.A. and Boes D.C. (1974) : Introduction to the Theory of Statistics, McGraw Hill.			
2. Freund J.E. (2001) : Mathematical Statistics, Prentice hall of India.			
3. Goon A.M., Gupta M.K. and Das Gupta B. (2014) : Fundamental of Statistics, Vol. I (5 th edition), World press, Calcutta.			
4. Rohatgi, V.K. and Saleh, A.K.M.E. (2001): An Introduction to Probability and Statistics, Second Edition, John Wiley & Sons.			
5. Rao C.R. (1973): Linear Statistical Inference and its Applications, Revised edition, Weley Eastern.			
6. Hoges J.L. and Lehmann E.L. (1964): Basic Concepts of Probability and Statistics, Holden Day.			
7. Snedecor G.W. and Cochran W.G. (1989) : Statistical Methods (8 th edition), Iowa State University Press.			
8. Sheskin, D. J. (2020). <i>Handbook of parametric and nonparametric statistical procedures</i> . Chapman and hall/CRC, Fifth Edition			
9. Roussas, G. G. (2015). <i>An introduction to probability and statistical inference</i> . Elsevier, Second Edition			
10. Almudevar, A. (2021). <i>Theory of statistical inference</i> . Chapman and Hall/CRC, First Edition			
11. Wasserman, L. (2013). <i>All of statistics: A concise course in statistical inference</i> , Second Edition			

Learning Outcomes: After completing this course, the students will be able to understand the followings; 1. About the point and interval estimation of the parameter(s) 2. Various properties of a good estimators and their theoretical development. 3. Concept of large sample, law for large number and related theorem. 4. Concept of small and large sample parametric test. 5. Concept and application of various non-parametric test.			
Course Title	STAMN42: Practical Based on STAMN41		
Paper will be based on numerical problems of Paper STAMN41			
Unit			Number of Practical
I			2
II			4
III, IV &V			24
Semester V			
Course Title: STAMJ51: Operations Research			
Category of Course: Major			
Credits& Hours of Teaching			
	Theory	Practical	Cumulative
Credits	4	1	5
Hours of Teaching (Total)	60	30	90
Course Objectives: 1. To understand optimization problems. 2. To develop linear programming problems (LPP) for real life situations. 3. To solve the LPP through different techniques under large number of decision variables and constraints. 4. To discuss the management of inventory problems. 5. To understand networking and project management.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Definitions and scope of operation research, different types of models in operations research – their construction and general method of solution.		7
II	Elements of linear programming problem (LPP): Canonical and standard forms, formulation of LPP, graphical method to solve two variable LPP, solution of LPP using simplex procedure, use of artificial variables in LPP, generation of extreme point solutions, principle of duality in LPP, statement and proof of duality theorem, simple problems based on duality theorem.		15
III	Transportation problem (TP): different methods of finding initial feasible solution of a TP, UV method of finding optimal solution of a TP, Problems of degeneracy, and special cases of TP. Assignment problem: Hungarian method to find optimal assignment, special cases of assignment problem.		15
IV	Inventory Control: Definitions of various costs involved in inventory control. Deterministic Economic Lot Size problems with and without shortages.		8
V	Theory of games: Two person zero-sum games, pure and mixed strategies, saddle point, maximin-minimax principle of rectangular games, games without saddle point, dominance and modified dominance property to reduce the game matrix and solution to rectangular game with mixed strategy, graphical solution of 2xN and Mx2 games, reduction of game problems to a LPP. Theory of Network – PERT, CPM.		15
Texts / References: 1. Taha, H. (2019). Operations Research: An Introduction, Pearson Education, 10th edition. 2. Hadley, G: (2002): Linear Programming, Narosa Publications. 3. Hillier, F.A and Lieberman, G.J. (2010): Introduction to Operations Research Concepts and cases, 9th Edition. 4. Gass S.I. (2011): Linear Programming- Methods and Applications, 5th edition, Dover Publications Inc. 5. McKinsey J.C.C. (2003): Introduction to the Theory and Games, Dover Publications Inc. 6. KantiSwaroop, Gupta P.K. and Singh M.M. (2007). Operations Research, 13th Edition, Sultan Chand and Sons.			

Learning Outcomes: After completing this course, a student will be able to: 1. Convert real life problem into LPP. 2. Utilize different approaches to solve the optimization problems. 3. Manage the problems of transportation, assignment, game for optimization purposes. 4. The management of inventory problems. 5. Learn project management and review techniques.			
Course Title: STAMJ52: Numerical Methods			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives: 1. To understand differencing methods to fit a polynomial. 2. To understand interpolation, extrapolation for missing or future observation. 3. Techniques of evaluating integral numerically. 4. Techniques for summation of mathematical series. 5. Solving deferential equations as well as linear and transcendental equations numerically.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	What is numerical analysis? Calculus of Finite differences. Operators: Δ , ∇ , δ , E and D and their relationships. factorial polynomials, separation of symbols, differences of zero. Taylor's series.		5
II	Interpolation and polynomial approximation Weierstarss theorem, extrapolation, Newton-Gregory's forward and backward interpolation formulae for equally-spaced arguments, divided differences and their properties, Newton's formula for divided difference, Lagrange's formula for unequal intervals, central difference formula due to Gauss, Stirling, Bessel, Laplace and Everett, remainder terms in interpolation formula. error analysis.		10
III	Inverse interpolation: Different methods of inverse interpolation. Numerical differentiation: Three-point formulae and five-point formulae Numerical Integration: General Quadrature formula: Trapezoidal rule, Simpson's one-third and three-eight rules, Weddle's rule, Newton-Cotes formula. Truncation error in the numerical integration formulae.		10
IV	Summation of series: Series whose general term is (i) the first difference of a function (ii) in geometric progression and (iii) a product of two functions. Difference equation: Linear difference equations (homogeneous and non-homogeneous) with constant coefficients.		10
V	Numerical solutions of differential equations: Euler's method, Milne's method, Picard's method and Runge-Kutta method. Method of solving linear and transcendental equations: Regula-falsi, bisection method, fixed-point iteration method, Newton-Raphson method, error analysis		10
Texts / References: 1. Scarborough, J. B. (2020). <i>Numerical Mathematical Analysis</i> (6th ed.). Oxford University Press and Oxford Book Co. 2. Freeman, H. (1961). <i>Finite Differences for Actuarial Students</i> (6th ed.). Cambridge University Press. 3. Atkinson, K., & Han, W. (2003). <i>Elementary Numerical Analysis</i> (3rd ed.). John Wiley & Sons, Inc. 4. Pettofrezzo, A. J. (2012). <i>Introductory Numerical Analysis</i> . United States: Dover Publications. 5. Hildebrand, F. B. (1987). <i>Introduction to Numerical Analysis</i> . India: Dover Publications. 6. Spiegel, M. R. (1971). <i>Schaum's outline of theory and problems of calculus of finite differences and difference equations</i> . United Kingdom: McGraw-Hill Education. 7. Conte, S. D., & de Boor, C. (1981). <i>Elementary Numerical Analysis: An Algorithmic Approach</i> (3rd ed.). McGraw-Hill.			

Learning Outcomes: After completing this course a student will learn to:			
1. To fit the polynomials of different degree using differencing operators.			
2. Interpolating and extrapolating the missing or future observations by Newton's, Langranze's and other techniques.			
3. Apply quadrature formula for evaluation of integrals numerically.			
4. Apply the methods of solving differential equations as well as linear and transcendental equations numerically.			
Course Title: STAMJ53: Statistical Inference -II			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	4	1	5
Hours of Teaching (Total)	60	30	90
Course Objectives:			
1. To introduce the concept and properties of a good estimator and their illustrations.			
2. To understand the meaning of lower bound, complete family of distributions.			
3. To introduce the idea about construction of minimum variance unbiased estimators through the use of sufficient statistics and their uniqueness.			
4. To illustrate various methods of point estimation of parameters along with their illustration.			
5. To illustrate various methods of interval estimation of parameters along with their illustration.			
6. To discuss the concept and importance of testing of hypothesis problem in detail.			
7. Elementary idea about the decision theoretic approach and Bayes estimation.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Properties of point estimators: unbiasedness, consistency of estimators and sufficient conditions for consistency, relative efficiency of an estimator, asymptotic efficiency, Cramer-Rao lower bound, sufficiency, factorization theorem (without proof), concept of complete sufficient statistics, minimum variance unbiased estimator, Rao-Blackwell and Lehmann-Scheffe Theorems.		15
II	Methods of point estimation : moments, maximum likelihood, minimum chi-square, least square with examples, CAN and BAN estimators, point estimates of location and scale parameters, regression coefficient, correlation coefficient and other useful parameters.		15
III	General method for constructing confidence intervals, shortest confidence intervals, confidence intervals for the parameters of univariate normal, two independent normal distributions and exponential distributions. error in the numerical integration formulae.		10
IV	Testing of statistical hypotheses, critical region, size and power of a test, most powerful test, randomized and non randomized test, Neyman Pearson lemma and its applications, Uniformly most powerful test, unbiased test, uniformly most powerful unbiased test, Likelihood ratio test, statement of its properties and applications.		15
V	Elementary idea about decision theory: Loss function, risk function, estimation and testing viewed as decision problems. Bayes rule.		5
Texts / References:			
1. Casella, G., & Berger, R. L. (2022). <i>Statistical Inference</i> (3rd ed.). Brooks/Cole, Cengage Learning.			
2. Mukhopadhyay, N. (2020). <i>Probability and statistical inference</i> . CRC Press.			
3. Kale, B.K. (1999): A First Course on Parametric Inference, Narosa Publishing House.			
4. Rohatgi, V.K. (1988): An Introduction to Probability and Mathematical Statistics, Wiley Eastern, New Delhi.			
5. Lehmann, E.L. (1986): Theory of Point Estimation, Student Edition.			
6. Lehmann, E.L.(1986): Testing Statistical Hypotheses, Student Editions.			
7. Rao, C.R. (1973): Linear Statistical Inference and its Applications, Wiley Eastern.			
8. Ferguson, T.S. (1967): Mathematical Statistics, Academic Press.			
9. Zacks, S. (1971): Theory of Statistical Inference, Wiley, New York.			
10. Sheskin, D. J. (2020). <i>Handbook of parametric and nonparametric statistical procedures</i> . Chapman and hall/CRC, Fifth Edition			
11. Roussas, G. G. (2015). <i>An introduction to probability and statistical inference</i> . Elsevier, Second Edition			
12. Almudevar, A. (2021). <i>Theory of statistical inference</i> . Chapman and Hall/CRC, First Edition			
13. Wasserman, L. (2013). All of statistics: A concise course in statistical inference, Second Edition			

Learning Outcomes: After completing this course, the students will be able to:			
1. Know that how the best estimators for the parameter(s) in any parametric family of distributions are chosen.			
2. Obtain the lower bound to the variance of unbiased estimators, complete sufficient statistics for the parameter.			
3. Construct the MVUE and UMVUE.			
4. Grasp various methods of point and interval estimation of parameters.			
5. Conceptualize the testing of hypothesis problem and related test.			
6. Understand decision theoretic approach and related estimation procedure.			
Course Title	STAMJ54: Practical Based on STAMJ51, STAMJ52 & STAMJ53		
Paper will be based on numerical problems of Paper STAMJ51, STAMJ52 & STAMJ53			
Paper& Unit			Hours of Practical
Paper STAMJ51unit I to V			30
Paper STAMJ52 unit I to V			30
Paper STAMJ53 unit I to V			30
Course Title	STAIN51: Internship		
The student will do summer internship work either in campus or outside of the campus.			
Internship Credit			Duration
2			Summer Vacation
Course Title: STAMV51: Structure and Visualization of Data			
Category of Course: Minor (Vocational)			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives			
1. Understand the principles of data structure and organization.			
2. Learn and apply key data visualization techniques.			
3. Develop proficiency in using visualization tools.			
4. Analyze data to extract meaningful insights and communicate findings.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Overview of data. scales of measurement, types variables. Data collection methods, and preparing data. Sources of data (surveys, web scraping, databases).Data structures: Arrays, lists, tables, and databases Data types and formats (CSV,SQL).Data normalization and schema design case studies of effective and ineffective visualizations.		15
II	Data cleaning exercise using Excel. Basic chart types: Bar, error bars, line, pie, and scatter plots. Explore advanced visualization methods. Heatmaps, histograms, box plots.		10
III	Numerical descriptors and their roles in data analysis for univariate and bivariate data. Measures of Inequalities		5
IV	Time series analysis basics, Plotting trends, seasonality, and anomalies, Tools for time series data visualization.		10
V	Explore visualization techniques for large datasets. Key performance indicators (KPIs) and business metrics, Creating executive dashboards Communicating insights to stakeholders.		5
Texts / References:			
1. Healy, K(2018). Data Visualization: A Practical Introduction". Princeton University Press.			
2. Wilke, C.O. (2019). Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures. Shroff/O'Reilly Publication.			
3. Ranganathan, K. (2023). Impactful Data Visualization: Hide and Seek with Graphs.			
Learning Outcomes:			

Students will be able to:			
1. Understand the importance and purpose of data visualization.			
2. Learn methods for collecting and preparing data.			
3. Explore how data is organized and structured.			
4. Get familiar with basic tools for data descriptions.			
Course Title	STAMV52: Practical Based on STAMV51		
Paper will be based on numerical problems of Paper STAMV51			
Unit			Hours of Practical
I			2
II			4
III			10
IV			8
V			6
Semester VI			
Course Title: STAMJ61: Stochastic Processes			
Category of Course: Major			
Credits& Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives:			
This course is designed to introduce:			
1. To understand stochastic processes			
2. To analyze discrete time Markov chains			
3. To understand Continuous-time Markov chains			
4. To understand Various counting processes and their applications & analysis.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Introduction to stochastic processes: Definition and examples of stochastic process, Classification of SPs according to state space and time domain. Types of stochastic processes: independent process, Bernoulli process, processes with independent increments, processes with stationary increments, Markov processes, counting processes, second-order processes, Gaussian processes, martingales, Brownian motion. Stationary process: weakly stationary and strongly stationary processes. Elementary problems and applications.		8
II	Markov chains: definition and examples of Markov chain (MC), transition probability matrix, order of Markov chain, Markov chain as graphs, higher transition probabilities, classification of states, basic limit theorem of Markov chain. Finite state Markov chains, Chapman-Kolmogorov equations, calculation of n-step transition probability and its limit. Classification of states, aperiodic & irreducible MC, recurrent & transient MC, concept of total probability with recursion, absorption probability, mean hitting time, mean return time, stationary distribution. Random walk and gambler's ruin problem.		13
III	Discrete state space continuous time MC: Kolmogorov forward and backward equations, jump probability, transition probability, generator matrix and its stationary distribution. Statistical inference in MC and Markov processes.		11
IV	Counting processes: Poisson process and related inter-arrival time distribution, superposition and decomposition of Poisson process, nonhomogeneous Poisson process. birth and death process, queuing model as a case of counting process.		13
Texts / References:			
1. Papoulis, A., &Unnikrishna Pillai, S. (2002). Probability, Random Variables, and Stochastic Processes (4th ed.). Tata McGraw-Hill.			
2. Ross, S. M. (2023). Introduction to Probability and Statistics for Engineers and Scientists (5th ed.). Academic			

Press. - Basu, A. K. (2005). Introduction to Stochastic Processes. Narosa Publishing. - Karlin, S., & Taylor, H. M. (1975). A First Course in Stochastic Processes, Vol. 1. Academic Press. - Castaneda, L. B., Arunachalam, A., & Dharmaraja, S. (2016). Introduction to Probability and Stochastic Processes with Applications. Wiley, Asian Edition. - Bharat, B. R. (2000). Stochastic Models: Analysis and Applications. New Age International, India. - Feller, W. (1968). Introduction to Probability and Its Applications, Vol. 1. Wiley Eastern. - Ross, S. M. (1996). Stochastic Processes. Wiley.
Learning Outcomes Students will be able to: - Define and classify various types of stochastic processes, including independent, Bernoulli, Markov, and Gaussian processes, and understand their properties such as stationary increments and martingales. - Comprehend the definition, transition probability matrix, and classification of states in Markov chains, and apply key concepts such as the Chapman-Kolmogorov equations, higher transition probabilities, and the basic limit theorem to practical problems. - Conceptualize discrete state space continuous time Markov chains, including Kolmogorov equations, jump probabilities, generator matrices, and their stationary distributions, and perform statistical inference on these processes. - Understand counting processes like the Poisson process, its inter-arrival time distribution, superposition and decomposition, and nonhomogeneous Poisson processes, and analyze birth and death processes and queuing models as specific cases of counting processes.

Course Title: STAMJ62: Statistical Process Control and Reliability			
Category of Course: Major			
Credits& Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives			
1. To introduce the fundamental problem-solving tool in Statistical Process Control (SPC), known as control charts 2. To determine whether a process is in control or if special causes of variation need to be addressed. 3. To discuss and demonstrate practical applications of basic control charts for variables and attributes, showing how they enable organizations to maintain and improve their processes. 4. To introduce the concepts of life testing experiments and reliability analysis. 5. To discuss estimation procedures for various life-time distributions in the presence of censored data. 6. To discuss various system configurations.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Statistical process and product control: Quality of a product, need for quality control, basic concept of process control, process capability and product control, general theory of control charts, causes of variation in quality, control limits, sub-grouping summary of out-of-control criteria.		8
II	Charts for attributes: p-chart, np-chart, c-chart, Charts for variables: R-chart, σ -charts. x-bar chart.		14
III	Life testing and reliability theory: Basic concepts of life testing experiments, reliability, hazard function, mean time to system failure and their inter-relationship. System reliability concepts: Parallel, series, series-parallel, parallel-series, bridge and k-out of- n systems. Concept of Redundancy and Preventive maintenance		8
IV	Elementary notion of censored data, type I and type II censoring schemes. Point and interval estimation procedures: exponential, Weibull, gamma, lognormal. UMVUE of reliability function and reliability hypothesis testing for exponential distribution. Stress-strength reliability and its estimation for the exponential distribution.		15
Texts / References:			
1. Montgomery, D. C. (2019). <i>Introduction to Statistical Quality Control</i> (8th ed.). Wiley. 2. Chakraborti, S., & Graham, M. A. (2019). <i>Nonparametric Statistical Process Control</i>. Wiley. 3. Mitra, A. (2021). <i>Fundamentals of quality control and improvement</i> (5th ed.). Wiley. 4. Wetherill, G. B., & Brown, D. W. (2013). <i>Statistical Process Control: Theory and Practice</i>. United States: Springer US. 5. Gross A. j. and Clark, V.A. (1975): <i>Survival Distribution: Reliability applications in the Biomedical Sciences</i>, John Wiley and Sons. 6. Elandt – Johnson, R.E. Johnson N. L.: <i>Survival Models and Data Analysis</i>, John Wiley and Sons. 7. Gertsbakh, I. B. (2000). <i>Reliability theory: with applications to preventive maintenance</i>. Springer Science & Business Media. 8. Balagurusamy E. (2017). <i>Reliability Engineering</i>, McGraw Hill Education India			
Learning Outcomes			
After study of this course students should be able to do the following:			
1. Understand chance and assignable causes of variability in a process 2. Explain the statistical basis of the Shewhart control chart, including choice of sample size, control limits, and sampling interval 3. Know how to design variables control charts such as X-bar and R charts, and attribute control charts such as p, np, c, u-charts. 4. Understand basic concepts of life testing experiments, reliability, and hazard function. 5. Learn about different censoring schemes such as Type I and Type II, and their applications 6. Know the different life-time distributions and related estimation problems.			

Course Title: STAMJ63: Econometrics and Actuarial Statistics			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	4	NIL	4
Hours of Teaching (Total)	60	NIL	60
Course Objectives: This course is designed to: 1. Understand econometric models and estimation techniques. 2. Address heteroscedasticity, autocorrelation, and multicollinearity. 3. Apply actuarial concepts in insurance and utility theory. 4. Calculate interest rates and annuities.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Nature of econometrics, the general linear model (GLM) and its extensions, ordinary least squares (OLS) estimation and prediction, generalized least squares (GLS) estimation and prediction, heteroscedastic disturbances, pure and mixed estimation.		13
II	Autocorrelation, its consequences and tests. Multicollinearity problem, its implications and tools for handling the problem, ridge regression. Instrumental variable estimation, errors in variables.		12
III	Expected value principle, utility and utility function, insurance and utility theory, mutually advantageous situation for insurance, survival function, time-until-death for a person, future life time, curtate future lifetime, force of mortality, relation of life table function to the survival function, Life table examples, deterministic survivorship group.		10
IV	Nominal and effective rates of interest and discount, force of interest and discount, compound interest, accumulation factor, insurance payable at the moment of death and at end of the year of death, concept of annuities, continuous and discrete life annuities, concept of premiums, continuous and discrete premiums.		10
Texts / References: 1. Promislow, S. D. (2014). <i>Fundamentals of Actuarial Mathematics</i> (3rd ed.). Wiley. 2. Deshmukh, S. R. (2009). <i>Actuarial Statistics: An Introduction Using R</i> (3rd ed.). 3. Borowiak, D. S., & Shapiro, A. F. (2014). <i>Financial and Actuarial Statistics: An Introduction</i> (2nd ed.). CRC Press. 4. Wooldridge, Jeffrey M. (2023). <i>Introductory Econometrics: A Modern Approach</i> (8th ed.). Cengage Learning. 5. Hayashi, F. (2000). <i>Econometrics</i> . Princeton University Press. 6. Greene, W. H. (2023). <i>Econometric Analysis</i> (8th ed.). Pearson. 7. Asteriou, D., & Hall, S. G. (2023). <i>Applied Econometrics</i> (5th ed.). Palgrave Macmillan. 8. Davidson, R., & MacKinnon, J. G. (2004). <i>Econometric Theory and Methods</i> (5th ed.). Oxford University Press. 9. Lo, A. W., & MacKinlay, A. C. (1999). <i>The Econometrics of Financial Markets</i> . Princeton University Press.			
Learning Outcomes: 1. Grasp the nature of econometrics and the application of general linear models (GLM), including ordinary least squares (OLS) and generalized least squares (GLS) for estimation and prediction. 2. Identify and manage problems such as heteroscedasticity, autocorrelation, and multicollinearity, employing techniques like ridge regression and instrumental variable estimation. 3. Comprehend the expected value principle, utility functions, and the application of insurance and utility theory, including the analysis of survival functions and life tables. 4. Master concepts of nominal and effective interest rates, force of interest, compound interest, and various types of annuities, as well as the computation of premiums for continuous and discrete life insurance products.			

Course Title: STAMJ64: Official and Vital Statistics			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives:			
1. To understand the official statistics, sources of different data required regarding economy, agriculture, demography, environment, social and cultural situations in a particular country.			
2. To understand vital statistics and their sources, trends and measures.			
3. To aware about population projections methods, family welfare programs and labour force statistics.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Introduction to Indian and International statistical systems. role, function and activities of central and state statistical organizations, organization of large-scale sample surveys, role of national sample survey organization general and special data dissemination systems.		10
II	System of collection of agricultural statistics, crop forecasting and estimation, productivity, fragmentation of holdings, support prices, buffer stocks, impact of irrigation projects. Statistics related to industries, foreign trade, balance of payment, cost of living, inflation, educational and other social statistics.		10
III	Some basic demographic concepts. Components of population change. Historical trends in population situation in the world. Brief description of Demographic transition theory. Sources of Demographic Data: Population census; Uses and limitations; Indian Censuses, Vital registration system, National Sample Surveys, Sample Registration System, Demographic Health Surveys (DHS) and other sample surveys.		10
IV	Measurement of Mortality and Life Table: Crude death rate, Standardized death rates, Age-specific death rates, Infant Mortality rate, Death rate by cause, Complete life table and its main features, Uses of life table. Maternal Mortality rate		5
V	Basic terms and concepts used in the study of fertility. Measurement of Fertility: Crude birth rate, general fertility rate, age specific birth rate, total fertility rate, gross reproduction rate, net reproduction rate. Concepts of population projections; population estimates and forecasts uses of population projections. Mathematical methods of measuring population projection. Population growth in developed and developing countries, evaluation and performance of family welfare programmes, projections of labour force and manpower.		10
Texts / References:			
1. Basic Statistics Relating to the Indian Economy (CSO), 1990.			
2. Guide to Official Statistics (CSO) 1999.			
3. Statistical System in India (CSO), 1995.			
4. Principles and accommodation of National Population Censuses, UNESCO.			
5. Panse, V.G.: Estimation of Crop Yields (FAO).			
6. Family Welfare Yearbook, Annual Publication of D/o Family Welfare.			
7. Monthly Statistics of Foreign Trade in India, DGCIS, Calcutta and other Govt. Publications.			
8. Ram Kumar (1986): Technical Demography, Wiley Eastern Ltd.			
9. Benjamin, B. (1969): Demographic Analysis, George, Allen and Unwin.			
10. Cox, P.R. (1970): Demography, Cambridge University Press.			
11. Spiegelman, M. (1969): Introduction to Demographic Analysis, Harvard University Press.			
12. Pathak, K. B. and Ram, F.(1992) Techniques of Demographic Analysis, Himalaya Publication House.			
Learning Outcomes: Upon completing this course, students will achieve the following learning outcomes:			
1. Students will learn national and international statistical systems for official statistics.			
2. They will learn to different sources of demographic data like census, vital registration systems, sample surveys.			
3. They will learn to measures of fertility, mortality and life-table techniques.			
4. Students will learn different population projection methods and their applications.			

Course Title	STAMJ65: Practical Based on STAMJ61, STAMJ62 &STAMJ64		
Paper will be based on numerical problems of Paper STAMJ61, STAMJ62 &STAMJ64			
Paper and Unit			Hours of Practical
Paper STAMJ61 Unit I to IV			30
Paper STAMJ62 Unit I to IV			30
Paper STAMJ64 Unit I to V			30
Semester VII			
Course Title: STAMJ71: Linear Models & Regression Analysis			
Category of Course: Major			
Credits& Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives:			
1. To understand the concepts of liner models and estimation of parameters.			
2. To learn g-inverse of a matrix and its application in estimation.			
3. Explain the applications of linear model in analysis of variance.			
4. To understand multiple comparison tests in comparison of means.			
5. To explore the idea of simple and multiple regression, residual analysis and detection of outliers.			
6. To understand the assumptions of regression analysis.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Gauss-Markov linear models, estimable functions, error and estimation space, normal equations and least square estimators, estimation of error variance, estimation with correlated observations, properties of least square estimators, generalized inverse of a matrix and solution of normal equations, variances and covariances of least square estimators.		15
II	One way and two-way classifications, fixed, random and mixed effects models. Analysis of variance (two-way classification only), Multiple comparison tests due to Tukey, Scheffe and Student-Newmann-Keuls.		5
III	Simple linear regression, multiple regression, fit of polynomials and use of orthogonal polynomials. Residuals and their plots as tests for departure from assumptions such as fitness of the model, normality, homogeneity of variances and detection of outliers and Remedies		15
IV	Multicollinearity, ridge regression, subset selection of explanatory variables, Mallow-Cp Statistics and its applications.		10
Texts / References:			
1. Goon, A.M., Gupta, M.K. and Das Gupta, B. (1967): An Outline of Statistical Theory, Vol. 2, The World Press Pvt. Ltd., Calcutta.			
2. Rao, C.R. (1973); Linear Statistical Inference and its Application, Wiley Eastern.			
3. Graybill, I.A. (1961): An Introduction to Linear Statistical Models, Vol. 1, McGraw Hill Book Co. Inc.			
4. Draper, N.R. and Smith H. (2014); Applied Regression Analysis, 4 th Ed. Wiley.			
5. Weisberg, S. (2005): Applied Linear Regression, 3 rd Ed. Wiley.			
6. Cook, R.D. and Weisberg, S. (2007): Residuals and Inference in Regression, 2 nd Ed. Chapman and Hall			
Learning Outcomes: Students will learn			
1. The estimation techniques and calculation of parameters of linear models.			
2. The concepts and applications of Fixed effects, random effects and mixed effects models.			
3. Properties of least square method f estimation of parameters of linear models.			
4. simple and multiple regression analysis, and computation of R ² and adjusted R ²			
5. Residual analysis and methods of obtaining residuals, computation of outliers.			
6. Diagnosis of multicollinearity, applications of ridge regression and Mallow-Cp Statistics.			

Course Title: STAMJ72: Measure & Probability Theory			
Category of Course: Major			
Credits& Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	Nil	3
Hours of Teaching (Total)	45	Nil	45
Course Objectives			
1. To understand the concepts of classes of sets, field, sigma field, limit s of sequence of sets.			
2. To learn probability measure and its properties.			
3. Understand measurable function, different convergence theorems.			
4. To understand characteristic function and different mode of convergence of sequence of random variables and their applications.			
5. To explore the idea of Law of large numbers, CLT and their applications.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Classes of sets, fields, sigma field, minimal sigma field, Borel sigma field, sequence of sets, limsup and liminf of a sequence of sets, measure, probability measure, properties of measure, Caratheodory extension theorem, Lebesgue and Lebesgue - Steiltzes measures.		15
II	Inverse function and their properties, measurable functions, random variables, sequence of random variables, Integration of a measurable function with respect to a measure, monotone convergence theorem, Fatou’s lemma, dominated convergence theorem.		10
III	Characteristic function, uniqueness theorem, Levy’s continuity theorem, Convergence in distribution, Convergence in probability, almost sure convergence.		10
IV	Borel-Cantelli lemma, independence of random variables, weak law and strong law of large numbers for the sequences of independently and identically distributed and non- i.i.d. random variables; Central limit theorem due Lindeberg- Levy for i.i.d. random variables, Liapounav's CLT for sequence of non- i.i.d. random variables.		10
Texts / References			
1. Robert, A. (1972): Real Analysis and Probability, Academic press.			
2. Billingsley, P. (2008): Probability and Measure, Third edition, Wiley.			
3. Dudley, R.M. (2002): Real Analysis and Probability; Cambridge Studies in Advanced Mathematics.			
4. Kingman, J.F.C. and Taylor, S.J. (2011): Introduction to Measure and Probability, Cambridge University Press.			
5. Basu, A. K. (2012): Measure Theory And Probability: Second Edition, Prentice Hall India Learning Private Limited.			
6. Athreya, K. B. and Lahiri, S. N. (2010): Measure Theory and Probability Theory Springer- Verlag New York Inc.			
Learning Outcomes: Students will learn			
1. Field, Sigma field, minimal field and Sigma field, Borel field and their properties, probability measure and their characterizations.			
2. Integration of measurable function with respect to a measure.			
3. Properties of different modes of convergence.			
4. Different modes of convergence and examples of convergence of sequence of random variables			
5. Continuity theorem and inversion theorem of characteristics function and their applications.			
6. Laws of large numbers and different types of CLT and their applications.			

Course Title: STAMJ73: Survey Sampling			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives			
1. To explain the concepts of fixed population vs super-population, and method of SRS and stratified sampling. 2. How to use auxiliary information in estimation 3. To explain the concept of unequal probability sampling 4. How to deal with non-response in surveys.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Fixed population and super-population approaches. Distinct features of finite population sampling, Probability sampling design and estimators along with basic statistical properties. Review of some important results in SRSWOR and SRSWR. Estimation of population mean/Total in stratified population, Allocation problem in stratified random sampling in case of fixed cost and also for specified precision. Expression for variance of stratified sample mean in case of fixed cost, Post stratification, Double sampling with post stratification, Deep stratification, Controlled sampling.		11
II	Ratio, product and regression methods of estimation. Estimation of population mean, evaluation of bias and variance/MSE to the first order of approximation, comparison with simple random sampling. Double sampling scheme, Some double sampling estimators for mean using auxiliary character (Ratio, regression and product) method of estimation, Some unbiased ratio type estimators for population mean.		11
III	Unequal probability sampling: PPSWR/WOR methods (including Lahiri’s scheme) and Des Raj estimator, Murthy estimator (for n=2). Horvitz Thompson Estimator of finite population total/mean, Expression for Variance (HTE) and its unbiased estimator, Issue of non-negative variance estimation, IIPS Scheme of sampling due to Midzuno-Sen.		13
IV	Concept of cluster sampling, two stage sampling, Non-sampling error with special reference to non-response problems.		10
Texts / References			
1. Cassel, C.M., Sarndal, C.E. and Wretman (1977): Foundation of Inference in Survey Sampling, Wiley Inter Science, New York. 2. Hansen, M. H., Hurwitz, W. N., & Madow, W. G. (1953). Sample survey methods and theory. 3. Singh, R., & Mangat, N. S. (2013). Elements of survey sampling (Vol. 15). 4. Springer Science & Business Media. 5. Murthy, M.N. (1977): Sampling Theory and Methods. Statistical Publication Society, Calcutta. 6. Mukhopadhyay, P. (1996): Inferential Problems in Survey Sampling, New Age International (P). 7. Sampath S. (2000) : Sampling Theory and Methods, Narosa Publishing House, New Delhi. 8. Sukhatme P.V., Sukhatme B.V., Sukhatme S. and Asok C. (1998): Sampling Theory of Surveys with Applications (4th ed.), Indian Society of Agricultural Statistics, New Delhi. 9. Goon A.M., Gupta M.K. and Das Gupta B. (2014): Fundamentals of Statistics, Vol. II, world Press, Calcutta. 10. Cochran, W.G.: Sampling Techniques (3rd edition. 1977), Wiley. 11. Singh D. and Chaudhary, F.S. (1986): Theory and Analysis of Sample Survey Designs, New Age International Publishers. 12. Singh, S. (2003) : Advanced sampling theory with applications. Springer-Science +Business Media, B.V. 13. Wu, C. and Thompson, M. E. (2020) : Sampling theory and practice. Springer.			
Learning Outcomes: Upon completing this course, students will achieve the following learning outcomes			
1. To select sampling units under probability and non-probability sampling procedures. 2. Constructing improved estimators using auxiliary information. 3. Student will be able to differentiate between the situations of equal probability and PPS sampling methods 4. How to tackle non-response problems in surveys.			

Course Title: STAMJ74: Advanced Statistical Inference			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	4	1	5
Hours of Teaching (Total)	60	30	90
Course Objectives 1. To understand Extend Cramer-Rao inequality and bounds. 2. To understand properties different statistics. 3. To understand maximum asymptotic properties of estimators. 4. To explore advanced inferential procedures.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Extension of Cramer-Rao inequality for multi-parameter case, Bhattacharya bounds.		10
II	Ideas of sufficient, minimal complete-sufficient statistics, sufficiency when the range of variable depends on a parameter, minimum variance unbiased estimators, Rao-Blackwell, Lehman-Scheffe and Basu theorem.		15
III	MLE and its asymptotic properties, Interval estimation by confidence sets, Neyman theory, uniformly most accurate intervals.		15
IV	Neyman-Pearson lemma, generalized Neyman-Pearson lemma, monotone likelihood ratio families, UMP tests for one and two sided alternatives, admissibility and unbiasedness of tests, type A and type A1 tests, similar tests, tests having Neyman structure, likelihood ratio test (LRT) asymptotic distribution of LRT statistic.		20
Texts / References 1. Kale, B.K. (1999): A First Course on Parametric Inference, Narosa Publishing House. 2. Rohatgi, V.K. (1988): An Introduction to Probability and Mathematical Statistics, Wiley Eastern, New Delhi. 3. Lehmann, E.L. (1986): Theory of Point Estimation, Student Edition. 4. Lehmann, E.L.(1986): Testing Statistical Hypotheses, Student Editions. 5. Zacks, S. (1971): Theory of Statistical Inference, Wiley, New York. 6. Sheskin, D. J. (2020). Handbook of parametric and nonparametric statistical procedures. Chapman and hall/CRC, Fifth Edition 7. Roussas, G. G. (2015). An introduction to probability and statistical inference. Elsevier, Second Edition 8. Almudevar, A. (2021). Theory of statistical inference. Chapman and Hall/CRC, First Edition 9. Wasserman, L. (2013). All of statistics: A concise course in statistical inference, Second Edition 10. Casella, G., & Berger, R. L. (2022). Statistical Inference (3rd ed.). Brooks/Cole, Cengage Learning. 11. Mukhopadhyay, N. (2020). Probability and statistical inference. CRC Press. 12. Niewiadomska-Bugaj, M., & Bartoszynski, R. (2020). Probability and statistical inference. John Wiley & Sons. 13. Silvey, S. D. (2017). Statistical inference. Routledge.			
Learning Outcomes: 1. Proficiency in extending Cramer-Rao inequality to compute the lower bound of the covariance matrix of unbiased estimators and understanding the Bhattacharya bound which provides tighter variance bounds where Cramer-Rao bounds may not be sufficient. 2. Ability to identify and apply sufficient, minimal complete-sufficient statistics, derive and utilize minimum variance unbiased estimators. 3. Understanding of MLE and its asymptotic properties. 4. Skill in constructing and interpreting confidence intervals using Neyman theory. 5. Capability to apply Neyman-Pearson lemma and generalized lemma for hypothesis testing. 6. Expertise in conducting UMP tests, evaluating test admissibility and unbiasedness, and performing LRT with an understanding of their asymptotic distributions.			

Course Title	STAMJ75: Practical Based on STAMJ71		
Paper will be based on numerical problems of Paper STAMJ71, STAMJ73, STAMJ74 as per following distribution			
Unit			Number of Practical
I			4
II			10
III			10
IV			6
Course Title	STAMJ76: Practical Based on STAMJ73& STAMJ74		
Paper will be based on numerical problems of Paper STAMJ73& STAMJ74 as per following distribution			
Paper STAMJ73 Unit I to III			30
Paper STAMJ74 Unit I to III			30
Course Title: STAMN71: Linear Models & Regression Analysis			
Category of Course: Minor			
Credits& Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives:			
1. To understand the concepts of liner models and estimation of parameters.			
2. To learn g-inverse of a matrix and its application in estimation.			
3. Explain the applications of linear model in analysis of variance.			
4. To understand multiple comparison tests in comparison of means.			
5. To explore the idea of simple and multiple regression, residual analysis and detection of outliers.			
6. To understand the assumptions of regression analysis.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Gauss-Markov linear models, estimable functions, error and estimation space, normal equations and least square estimators, estimation of error variance, estimation with correlated observations, properties of least square estimators, generalized inverse of a matrix and solution of normal equations, variances and covariances of least square estimators.		15
II	One way and two-way classifications, fixed, random and mixed effects models. Analysis of variance (two-way classification only), Multiple comparison tests due to Tukey, Scheffe and Student-Newmann-Keuls.		5
III	Simple linear regression, multiple regression, fit of polynomials and use of orthogonal polynomials. Residuals and their plots as tests for departure from assumptions such as fitness of the model, normality, homogeneity of variances and detection of outliers and Remedies		15
IV	Multicollinearity, ridge regression, subset selection of explanatory variables, Mallow-Cp Statistics and its applications.		10
Texts / References:			
1. Goon, A.M., Gupta, M.K. and Das Gupta, B. (1967): An Outline of Statistical Theory, Vol. 2, The World Press Pvt. Ltd., Calcutta.			
2. Rao, C.R. (1973); Linear Statistical Inference and its Application, Wiley Eastern.			
3. Graybill, I.A. (1961): An Introduction to Linear Statistical Models, Vol. 1, McGraw Hill Book Co. Inc.			
4. Draper, N.R. and Smith H. (2014); Applied Regression Analysis, 4th Ed. Wiley.			
5. Weisberg, S. (2005): Applied Linear Regression, 3rd Ed. Wiley.			
6. Cook, R.D. and Weisberg, S. (2007): Residuals and Inference in Regression, 2 nd Ed. Chapman and Hall			
Learning Outcomes: Students will learn			
1. the estimation techniques and calculation of parameters of linear models.			

2. the concepts and applications of Fixed effects, random effects and mixed effects models.			
3. properties of least square method f estimation of parameters of linear models.			
4. simple and multiple regression analysis, and computation of R ² and adjusted R ²			
5. residual analysis and methods of obtaining residuals, computation of outliers.			
6. diagnosis of multicollinearity, applications of ridge regression and Mallow-Cp Statistics.			
Course Title: STAMN72: Practical Based on STAMN71			
Paper will be based on numerical problems of Paper STAMN71			
Unit			Hours of Practical
I			4
II			10
III			10
IV			6
Semester VIII			
Course Title: STAMJ81: Design and Analysis of Experiments			
Category of Course: Major			
Credits& Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives			
1. To understand the concepts of basic designs and analysis in the case of missing observations.			
2. To learn Analysis of Covariance (ANCOVA) for CRD and RBD under one concomitant variable.			
3. Understand the concept of Incomplete Block Designs, BIBD and PBIBD.			
4. To understand factorial experiments and analysis, nested designs and fractional factorial.			
5. To explore the idea Split plot and Strip Plot designs.			
6. To understand application response area, clinical trials, RCT.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Review of linear estimation and basic designs, missing plot technique: General theory and applications, Analysis of Co-variance for CRD and RBD.		10
II	Incomplete block design: Balanced incomplete block designs, simple lattice designs, Two-associate partially balanced incomplete block designs: association scheme and intra block analysis, group divisible designs.		10
III	General factorial experiments, factorial effects; estimates and testing the significance of factorial effects; study of 2 ⁿ and 3 ^r factorial experiments in randomized blocks; complete and partial confounding, construction of symmetrical confounded factorial experiments, fractional replications for symmetrical factorials, split plot and strip-plot experiments.		15
IV	Application areas: Response surface experiments; first order designs, and orthogonal designs; clinical trials, treatment-control designs; model variation and use of transformation; Tukey’s test for additivity.		10
Texts / References			
1. Dean, A., & Voss, D. (2023). <i>Design and Analysis of Experiments</i> (7th ed.). Springer.			
2. Montgomery, C. D. (2023). <i>Design and Analysis of Experiment</i> (9th ed.). Wiley, New York.			
3. Myers, R. H. (2016). <i>Response Surface Methodology</i> (4th ed.). John Wiley & Sons.			
4. Dey, A. (1986). <i>Theory of Block Designs</i> . Wiley Eastern.			
5. Das, M., & Giri, N. (1979). <i>Design and Analysis of Experiments</i> . Wiley Eastern.			
6. Joshi, D. D. (1987). <i>Linear Estimation and Design of Experiments</i> . Wiley Eastern.			
7. Giri, N. (1986). <i>Analysis of Variance</i> . South Asian Publishers.			

Learning Outcomes: Students will learn			
1. Basic fundamentals of designs, complete block designs their layout and analysis under missing observations.			
2. Incomplete block designs their types, analysis and applications.			
3. properties of different convergence theorems.			
4. 2 ⁿ , 3 ⁿ factorial experiments and their analysis, obtain confounded effects under different replications.			
5. Applications of Split plot (Nested designs), Strip-plot designs.			
6. Treatment control designs and Randomized control trails with applications.			
Course Title: STAMJ82: Multivariate Analysis			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives			
1. To understand the basic concepts of vector space and matrices.			
2. To impart knowledge about multivariate normal and its derived distributions.			
3. Estimation and hypothesis testing for the parameters of multivariate normal populations.			
4. To understand different dimension reduction methods.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Overview of multivariate data, multivariate normal distribution and its properties. Random sampling from multivariate normal distribution. Maximum likelihood estimators of parameters, distribution of sample mean vector, testing of mean vector for one and two sample problems.		15
II	Wishart matrix – its distribution and properties, distribution of sample generalized variance, testing of equality of covariance matrices. Overview of multiple and partial correlation coefficients, null and non-null distribution of multiple correlation coefficients.		8
III	Hotelling’s T ² statistic – its distribution and properties, applications in testing of mean vector for one and more multivariate normal population and on equality of components of mean vector in two multivariate normal populations. Mahalanobis D ² statistic and its distribution.		7
IV	Discrimination & classification, procedure of classification based on several multivariate populations with emphasis on multivariate normal populations, probability of misclassification, idea of sample discriminant function.		8
V	Idea of Principal components analysis, Canonical correlation analysis with their applications.		7
Texts / References			
1. Johnson, R. A., & Wichern, D. W. (2015). <i>Applied Multivariate Statistical Analysis</i> (6th ed.). Pearson.			
2. Anderson, T. W. (2003). <i>An Introduction to Multivariate Statistical Analysis</i> (3rd ed.). Wiley.			
3. Morrison, D. F. (2004). <i>Multivariate Statistical Methods</i> (4th ed.). Brooks/Cole.			
4. Muirhead, R. J. (1982). <i>Aspects of Multivariate Statistical Theory</i> . Wiley.			
5. Carroll, J. D., & Green, P. E. (2002). <i>Analyzing Multivariate Data</i> . Cengage Learning.			
6. Hair, J. F., Black, W. C., Anderson, R. E., & Tatham, R. L. (2006). <i>Multivariate Data Analysis</i> (7th ed.). Pearson.			
Learning Outcomes: After completing this course a student will have knowledge of			
1. Applications of multivariate normal distribution.			
2. Sampling distributions of estimators of parameters of MVN distribution.			
3. Comparison of MVN populations.			
4. Classification of observations in two or more populations.			
5. Various dimension reduction techniques like Principal Component Analysis, Factor Analysis and Canonical Correlation Analysis.			

Course Title: STAMJ83: Bayesian Inference			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives			
1. Provide an overview of statistical decision theory. 2. To equip students with fundamentals of Bayesian theory. 3. Develop skills in Bayesian inference. 4. Introduce basic Bayesian computation techniques and their practical applications.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	An overview of Statistical Decision Theory, decision problem and 2-person game, loss functions, risk functions, decision rules and utility functions. Concept of admissibility, minimaxity and completeness, Bayes rules, admissibility of Bayes rules.		14
II	Different types of informative and non-informative prior distributions: conjugate priors, Jeffreys’ priors, maximum entropy priors, invariant priors. Summarizing posterior distributions and the posterior as an average of the prior and data.		12
III	Bayes point and interval estimation, comparisons with likelihood based inferences; inferences for standard parameter models. Concepts of prior odds, posterior odds, Bayes factors. Bayes testing for simple and composite hypotheses. Lindley’s paradox, Bayesian model checking and comparison, Predictive distribution. Brief Idea of point and interval prediction.		14
IV	Elementary idea of Bayesian computation.		5
Texts / References			
1. Berger, J.O. (1985): Statistical Decision Theory and Bayesian Analysis, Springer Verlag. 2. Robert, C.P. and Casella, G. (2004): Monte Carlo Statistical Methods, Springer Verlag. 3. Leonard, T., & Hsu, J. S. (2001). Bayesian methods: an analysis for statisticians and interdisciplinary researchers (Vol. 5). Cambridge University Press. 4. Bernardo, J. M., & Smith, A. F. (2000). Bayesian theory. John Wiley & Sons. 5. Robert, C. P. (2007). The Bayesian choice: from decision-theoretic foundations to computational implementation (Vol. 2). New York: Springer. 6. Gamerman, D., & Lopes, H. F. (2006). Markov chain Monte Carlo: stochastic simulation for Bayesian inference. Chapman and Hall/CRC. 7. Box, G.P. and Tiao, G.C(1992): Bayesian Inference in Statistical Analysis, Addison-Wesley. 8. Gelman, A., Carlin, J. B., Stern, H. S., & Rubin, D. B. (1995). Bayesian data analysis. Chapman and Hall/CRC. 9. Bansal, A.K. (2007). Bayesian Parametric Inference. Alpha Science International Ltd. 10. Sinha, S.K. (1998). Bayesian Estimation. New Age International Pvt. Ltd.			
Learning Outcomes: Upon completing this course, students will achieve the following learning outcomes			
1. Key concepts, including decision problems, loss functions, risk functions, and utility functions. 2. Develop knowledge of prior, posterior distributions and prior elicitation. 3. Parameter estimation under different loss functions and testing of hypothesis. 4. Conduct Bayes estimation for the real-world problems and compare these methods with likelihood-based inferences.			

Course Title: STAMJ84: Statistical Aspects of Machine Learning			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives This course is designed to introduce: 1. Proficiency in python for machine learning regarding statistical modeling 2. Apply supervised and unsupervised learning techniques for multivariate analysis 3. Perform data preprocessing and exploratory statistical analysis 4. Understand and implement neural networks			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Introduction to Python, Overview of machine learning concepts and applications: Supervised, unsupervised, and reinforcement learning. Types of learning algorithms: regression, classification, and clustering.		5
II	Data Preprocessing and Visualization: Data cleaning and handling missing values. Feature scaling and normalization. Handling categorical variables. Exploratory data analysis and visualization using Python libraries (e.g., Pandas, Matplotlib, Seaborn).		8
III	Regression Analysis: Linear regression and multiple linear regression. Polynomial regression and feature transformation. Regularization techniques (L1, L2). Evaluation metrics for regression models. Classification Methods: Logistic regression and probabilistic models. Naive Bayes classifiers. Decision trees and ensemble methods (random forests, boosting). Support vector machines for classification. Performance evaluation and metrics for classification models.		12
IV	Unsupervised Learning Techniques: Clustering algorithms (K-means, hierarchical clustering). Dimensionality reduction techniques (PCA, t-SNE). Evaluation and validation of clustering results.		10
V	Introduction to Neural Networks: Basics of artificial neural networks (ANNs). Activation functions and training algorithms. Feedforward neural networks. Backpropagation algorithm.		10
Texts / References 1. Hastie, T., Tibshirani, R., & Friedman, J. H. (2009). <i>The Elements of Statistical Learning: Data Mining, Inference, and Prediction</i> (2nd ed.). Springer. 2. Géron, A. (2019). <i>Hands-On Machine Learning with Scikit-Learn, Keras, &TensorFlow: Concepts, Tools, and Techniques for Building Intelligent Systems</i> (2nd ed.). O'Reilly Media. 3. Müller, A. C., & Guido, S. (2016). <i>Introduction to Machine Learning with Python: A Guide for Data Scientists</i>. O'Reilly Media. 4. Raschka, S. (2017). <i>Python Machine Learning: Machine Learning in Python: A Practical Introduction with Scikit-learn</i> (2nd ed.). Packt Publishing. 5. Aurélien, G. (2017). <i>Deep Learning with Python</i>. Packt Publishing. 6. Chollet, F. (2018). <i>Deep Learning with Keras</i>. Manning Publications. 7. Liu, J., Tang, X., & Liu, H. (2019). <i>Practical Machine Learning: A Python-Based Guide</i>. O'Reilly Media 8. John M. Zelle, Python Programming: An Introduction to Computer Science, 2004. 9. Chun W. J. (2010), Core Python Programming, Pearson. 10. Murphy, K. P. (2012). Machine learning: A probabilistic perspective. MIT Press. 11. Goodfellow, I., Bengio, Y., &Courville, A. (2016). Deep learning. MIT Press.			
Learning Outcomes: Students will be able to 1. Develop a strong foundation in Python programming and its libraries for machine learning, including data preprocessing, visualization, and handling various data types. 2. Implement and evaluate machine learning algorithms for regression, classification, and clustering, understanding their applications and limitations. 3. Conduct data cleaning, feature scaling, normalization, and exploratory data analysis using Python tools like Pandas, Matplotlib, and Seaborn. 4. Grasp the basics of artificial neural networks, including activation functions, training algorithms, and			

backpropagation, and apply them to real-world problems.			
Course Title: STAMJ85: Practical Based on STAMJ81			
Paper will be based on numerical problems of Paper STAMJ81			
Paper & Unit			Hours of Practical
I			4
II			6
III&IV			20
Course Title: STAMJ86: Practical Based on STAMJ82, STAMJ83& STAMJ84.			
Paper will be based on numerical problems of Paper STAMJ82, STAMJ83& STAMJ84			
Paper & Unit			Hours of Practical
STAMJ82 Unit I to V			30
STAMJ83 Unit I to IV			30
STAMJ84 Unit I to V			30
Course Title: STAMN81: Design and Analysis of Experiments			
Category of Course: Minor			
Credits& Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives			
1. To understand the concepts of basic designs and analysis in the case of missing observations.			
2. To learn Analysis of Covariance (ANCOVA) for CRD and RBD under one concomitant variable.			
3. Understand the concept of Incomplete Block Designs, BIBD and PBIBD.			
4. To understand factorial experiments and analysis, nested designs and fractional factorial.			
5. To explore the idea Split plot and Strip Plot designs.			
6. To understand application response area, clinical trials, RCT.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Review of linear estimation and basic designs, missing plot technique: General theory and applications, Analysis of Co-variance for CRD and RBD.		10
II	Incomplete block design: Balanced incomplete block designs, simple lattice designs, Two-associate partially balanced incomplete block designs: association scheme and intra block analysis, group divisible designs.		10
III	General factorial experiments, factorial effects; estimates and testing the significance of factorial effects; study of 2 ⁿ and 3 ^r factorial experiments in randomized blocks; complete and partial confounding, construction of symmetrical confounded factorial experiments, fractional replications for symmetrical factorials, split plot and strip-plot experiments.		15
IV	Application areas: Response surface experiments; first order designs, and orthogonal designs; clinical trials, treatment-control designs; model variation and use of transformation; Tukey's test for additivity.		10
Texts / References			
1. Dean, A., & Voss, D. (2023). <i>Design and Analysis of Experiments</i> (7th ed.). Springer.			
2. Montgomery, C. D. (2023). <i>Design and Analysis of Experiment</i> (9th ed.). Wiley, New York.			
3. Myers, R. H. (2016). <i>Response Surface Methodology</i> (4th ed.). John Wiley & Sons.			
4. Dey, A. (1986). <i>Theory of Block Designs</i> . Wiley Eastern.			
5. Das, M., & Giri, N. (1979). <i>Design and Analysis of Experiments</i> . Wiley Eastern.			
6. Joshi, D. D. (1987). <i>Linear Estimation and Design of Experiments</i> . Wiley Eastern.			
7. Giri, N. (1986). <i>Analysis of Variance</i> . South Asian Publishers.			
Learning Outcomes: Students will learn			
1. Basic fundamentals of different designs.			
2. Incomplete block designs their types, analysis and applications.			
3. properties of different convergence theorems.			

4. 2^n , 3^n factorial experiments and their analysis,	
5. Applications of designs.	
Course Title: STAMN82: Practical Based on STAMN81	
Paper will be based on numerical problems of Paper STAMN81	
Unit	Hours of Practical
I	4
II	6
III & IV	20

Semester VIIIR			
Course Title: STAMJ8R1: Distribution Theory			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives			
1. To introduce the concept univariate and bivariate probability distributions. 2. To develop various distributions through various approaches. 3. To develop concept of mixture and truncated probability distributions. 4. To understand different probability-inequalities. 5. To understand non-central sampling distributions. 6. To conceptualize order statistics and their functions.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Brief review of basic distribution theory, joint, marginal conditional pmf and pdf, standard discrete and continuous distributions, bivariate normal, bivariate exponential, multivariate normal and multinomial distributions, functions of random variables and their distributions using Jacobian of transformation and other tools.		15
II	Compound, truncated and mixture distributions, multiple and partial correlations, linear and multiple regressions. Markov, Holder, Jensen, Liapunov inequalities.		10
III	Sampling distributions, non-central chi-square, t and F distributions and their properties. Distributions of quadratic forms under normality and related distribution theory.		10
IV	Order statistics, their distributions and properties, joint and marginal distributions of order statistics, moments of order statistics and recurrence relations, extreme values and their asymptotic distributions with applications. approximating distributions Δ -Method & its applications, approximating distributions of sample moments, transformations of statistics.		10
Texts / References			
1. Dudewicz, E.J. and Mishra, S.N. (1988): Modern Mathematics Statistics, Wiley International students edition. 2. Rohatagi, V.K. (1984): An introduction to Probability Theory and Mathematical Statistics, Wiley Eastern. 3. Rao, C.R. (1973); Linear Statistical Inference and its Applications, Wiley Eastern. 4. Pitman, J. (1993): Probability, Narosa Publishing House. 5. Jonson, S. and Kotz, S. (1972): Distribution in Statistics Vol. I-II & III, Houghton and Mifflin. 6. Mukhopadhyay, P. (1996): Mathematical Statistics, NCBA. 7. Ferguson, T.S. (1967): Mathematical Statistics, Academic Press. 8. Forbes, C., Evans, M., Hastings, N., & Peacock, B. (2011). <i>Statistical distributions</i> (1st ed.). John Wiley & Sons. 9. Hogg, R. V., McKean, J. W., & Craig, A. T. (2021). <i>Introduction to mathematical statistics</i> (8th ed.). Pearson Education India. 10. Feller, W., & Morse, P. M. (1958). <i>An introduction to probability theory and its applications</i>.			
Learning Outcomes:After completing this course, the students will learn the followings;			
1. Development of univariate and bivariate probability distributions through various approaches. 2. Different transformation techniques for obtaining the distribution of function one or more random variables. 3. Importance of mixture and truncated probability distributions.			

4. To understand the applicability of different probability inequalities and their development.			
5. Development of non-central sampling distributions viz., chi, t and F.			
6. Importance of order statistics, distribution of different order statistics and related functions.			
7. Moments of order statistics and their asymptotic distributions.			
Course Title: STAMJ8R2: Technical Demography			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	3
Hours of Teaching (Total)	45	30	75
Course Objectives			
1. To understand the difference between demography and Population studies and sources of collecting demographic data.			
2. To understand quality of data and concept of population transition theory.			
3. Understand the concept of measures of fertility, models of reproduction and different types of birth interval data.			
4. To understand concepts of mortality its measure, Life table types and its construction methods			
5. To explore the idea of migration, migration models and its measures.			
6. To understand population projections, nuptiality and its measure.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Coverage and content errors in demographic data, Chandrasekharan—Deming formula to check completeness of registration data, adjustment of age data- use of Whipple, Myer and UN indices. Population transition theory.		10
II	Measures of fertility; stochastic models for reproduction, distributions of time of birth, inter-live birth intervals and of number of births (for both homogeneous and homogeneous groups of women), estimation of parameters; estimation of parity progression from open birth interval data.		10
III	Measures of Mortality; construction of abridged life tables, infant mortality rate and its adjustments, model life table.Stable and quasi-stable populations, intrinsic growth rate. Models of population growth and their filling to population data.		10
IV	Internal migration and its measurement, migration models, concept of international migration.Methods for population projection, component method of population projection, Nuptiality and its measurements.		15
Texts / References			
1. Kumar, R. (1986): Technical Demography, Wiley Eastern Ltd.			
2. Benjamin, B. (1969): Demographic Analysis, George, Allen and Unwin.			
3. Chiang, C.L. (1968): Introduction to Stochastic Progression.			
4. Cox, P.R. (1970): Demography, Cambridge University Press.			
5. Keyfitz, N. (1977): Introduction to the Mathematics of Population-with Revisions, Addison-Wesley, London.			
6. Spiegelman, M. (1969): Introduction to Demographic Analysis, Harvard University Press.			
7. Wolfenden, H.H. (1954): Population Statistics and Their Compilation, Am Actuarial Society.			
Learning Outcomes: Students will learn			
1. Basics of demography and its sources and quality of demographic data.			
2. Fertility models, stochastic models, reproduction modeling, PPR and birth intervals data.			
3. Various measures of mortality and life table techniques.			
4. Population growth, projections, and migration.			

Course Title: STAMJ83:Bayesian Inference			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives 1. Provide an overview of statistical decision theory. 2. To equip students with fundamentals of Bayesian theory. 3. Develop skills in Bayes point and interval estimation and in performing Bayesian hypothesis testing. 4. Introduce basic Bayesian computation techniques and their practical applications.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	An overview of Statistical Decision Theory, decision problem and 2-person game, loss functions, risk functions, decision rules and utility functions. Concept of admissibility, minimaxity and completeness, Bayes rules, admissibility of Bayes rules.		14
II	Different types of informative and non-informative prior distributions: conjugate priors, Jeffreys' priors, maximum entropy priors, invariant priors. Summarizing posterior distributions and the posterior as an average of the prior and data.		12
III	Bayes point and interval estimation, comparisons with likelihood based inferences; inferences for standard parameter models. Concepts of prior odds, posterior odds, Bayes factors. Bayes testing for simple and composite hypotheses. Lindley's paradox, Bayesian model checking and comparison, Predictive distribution. Brief Idea of point and interval prediction.		14
IV	Elementary idea of Bayesian computation.		5
Texts / References 1. Berger, J.O. (1985): Statistical Decision Theory and Bayesian Analysis, Springer Verlag. 2. Robert, C.P. and Casella, G. (2004): Monte Carlo Statistical Methods, Springer Verlag. 3. Leonard, T., & Hsu, J. S. (2001). Bayesian methods: an analysis for statisticians and interdisciplinary researchers (Vol. 5). Cambridge University Press. 4. Bernardo, J. M., & Smith, A. F. (2000). Bayesian theory. John Wiley & Sons. 5. Robert, C. P. (2007). The Bayesian choice: from decision-theoretic foundations to computational implementation (Vol. 2). New York: Springer. 6. Gamerman, D., & Lopes, H. F. (2006). Markov chain Monte Carlo: stochastic simulation for Bayesian inference. Chapman and Hall/CRC. 7. Box, G.P. and Tiao, G.C(1992): Bayesian Inference in Statistical Analysis, Addison-Wesley. 8. Gelman, A., Carlin, J. B., Stern, H. S., & Rubin, D. B. (1995). Bayesian data analysis. Chapman and Hall/CRC. 9. Bansal, A.K. (2007). Bayesian Parametric Inference. Alpha Science International Ltd. 10. Sinha, S.K. (1998). Bayesian Estimation. New Age International Pvt. Ltd.			
Learning Outcomes: Upon completing this course, students will achieve the following learning outcomes 1. Key concepts, including decision problems, loss functions, risk functions, and utility functions. 2. Develop knowledge of prior, posterior distributions and prior elicitation. 3. Parameter estimation under different loss functions and testing of hypothesis. 4. Conduct Bayes estimation for the real-world problems and compare these methods with likelihood-based inferences.			
Course Title: STAMJ8R4 :Advanced Theory of Sample Surveys			
Category of Course: Major			
Credits & Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives 1. To explain the concepts of errors in sample surveys 2. To explain the of RRT 3. To understand the problem of small area estimation			

Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Non–Sampling Errors:Different kinds of non – sampling errors. Frame imperfections. Estimation in the presence of frame imperfections. Non–Response:Sources of non – response; unit and item non – response and their measures; Dealing with non – response : call backs and follow – ups; sub-sampling of non – respondents; imputation techniques for unit and item non – response.		11
II	Measurement Errors:Nature of measurement errors; models of measurement errors for estimating population parameters. Interpenetrating sub-samples.		11
III	Randomized Response Technique:Warner’s model; related and unrelated questionnaire models; use of binary responses.		13
IV	Issues in Small Area Estimation:Preliminaries; background of domain estimation; basic estimation methods for domains; direct estimators; synthetic estimators; generalized regression estimators.		10
Texts / References 1. Cassel, C.M., Sarndal, C.E. and Wretman (2013): Foundation of Inference in Survey Sampling, 2nd Ed., Wiley Inter Science, New York. 2. Murthy, M.N. (1977): Sampling Theory and Methods. Statistical Publication Society, Calcutta. 3. Mukhopadhyay, P. (1996): Inferential Problems in Survey Sampling, New Age International (P). 4. Sukhatme, P.V., Sukhatme, B.V., Sukhatme, S. and Asok, C. (1984): Sampling Theory of Surveys with Applications, Iowa State University Press and Indian Society of Agricultural Statistics. 5. Cochran, W.G.: Sampling Techniques (3rd edition. 1977), Wiley. 6. Singh D. and Chaudhary, F.S. (1986): Theory and Analysis of Sample Survey Designs, New Age International Publishers. 7. Chaudhuri, A. and R. Mukherjee (1988): Randomised response: theory and techniques, New York, Marcel Deckker Inc. 8. Singh, S. (2003) : Advanced sampling theory with applications. Springer-Science +Business Media, B.V. 9. Wu, C. and Thompson, M. E. (2020) : Sampling theory and practice. Springer.			
Learning Outcomes: Upon completing this course, students will achieve the following learning outcomes 5. How to tackle different types of errors in real life situations. 6. How to collect the information when it is of sensitive nature 7. How to get the estimates under the situation of small area.			
Course Title: STAMJ8R5 :Statistical Process and Quality Control			
Category of Course: Major			
Credits& Hours of Teaching			
	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching (Total)	45	30	75
Course Objectives 1. To introduce the fundamental problem-solving tool in Statistical Process Control (SPC), known as control charts 2. To determine whether a process is in control or if special causes of variation need to be addressed. 3. To discuss and demonstrate practical applications of basic control charts for variables and attributes, showing how they enable organizations to maintain and improve their processes. 4. To introduce single-, double-, and sequential-sampling plans as quality assurance tools. 5. To introduce various capability indices to estimate the process capability. 6. To discuss the use of the 2^{k-p} system of fractional factorial designs.			
Course Content: The course content be divided into minimum 3 Units to maximum 5 units along with defined Hours of teaching for each unit.			
Units	Course Content		Hours of Teaching
I	Basic concepts of process monitoring and control; process capability and process optimization.		6
II	General theory and review of control charts for attribute and variable data; O.C. and A.R.L. of control charts; control by gauging; moving average and exponentially weighted moving average charts; Cu-Sum charts using V-masks and decision intervals; Economic		16

	design of X-bar chart.	
III	Acceptance sampling plans for attributes inspection; single and double sampling plans and their properties; plans for inspection by variables for one-sided and two sided specification. Mil Std. and IS plans; continuous sampling plans of Dodge type and Wald-Wolfowitz type and then properties. Sequential sampling plan and its properties; Bayesian sampling plans.	9
IV	Capability indices C_p , C_{pk} and C_{pm} ; estimation, confidence intervals and tests of hypotheses relating to capability indices for normally distributed characteristics.	7
V	Use of design of experiments in SPC; factorial experiments, fractional factorial designs; construction of such designs and analysis of data. Multivariate quality control; use of control ellipsoid and of utility functions.	7

Texts / References

1. Montgomery, D.C. (2019): Introduction to Statistical Quality Control; 8th edition, Wiley.
2. Chakraborti, S. and Graham, M. A. (2019) Nonparametric Statistical Process Control, Wiley
3. Mitra, A. (2021). Fundamentals of quality control and improvement. 5th edition, Wiley
4. Wetherill, G. B., Brown, D. W. (2013). Statistical Process Control: Theory and Practice. United States: Springer US.
5. Oakland, J., Oakland, R. J. (2018). Statistical Process Control. United Kingdom: Taylor & Francis.
6. Aslam, M., Saghir, A., Ahmad, L. (2020). Introduction to Statistical Process Control. United Kingdom: Wiley.

Learning Outcomes: After careful study of this course students should be able to do the following

1. Understand chance and assignable causes of variability in a process
2. Explain the statistical basis of the Shewhart control chart, including choice of sample size, control limits, and sampling interval
3. Know how to design variables control charts such as $\bar{X}$ and R charts, EWMA and CUSUM charts, and attribute control charts such as p, np, c, u-charts.
4. Calculate and properly interpret process capability ratios, for example, C_p , C_{pk} and C_{pm} .
5. Understand the role of acceptance sampling in modern quality control systems and to know how single-, double-, and sequential-sampling plans are used.
6. Know how to use the 2^{k-p} system of fractional factorial designs.

Course Title: STAMJ8R6: Practical Based on STAMJ8R1, STAMJ8R3, STAMJ8R4 and STAMJ8R5 (student will opt any one)

Paper and Unit	Hours of practical
STAMJ8R all units	30
STAMJ8R2 all units	30
STAMJ8R3 all units	30
STAMJ8R4 all units	30
STAMJ8R5 all units	30

Course Title: STADS8R1: Project/Dissertation

Student will complete a project/ Dissertation on a topic as decided by the department. The mode of and area of dissertation will be different for each student. The dissertation will be of 12 credits.

Course Title: STAMN8R1: Elective paper

Any one out of STAMJ8R1 STAMJ8R2, STAMJ8R3, STAMJ8R4 and STAMJ8R5) Syllabus and credit will be same as mentioned against the papers.

Course Title: STAMN8R2: Practical paper

Any one out of STAMJ8R1 STAMJ8R2, STAMJ8R3, STAMJ8R4 and STAMJ8R5) Syllabus and credit will be same as mentioned against the papers.

Practical based on the elective paper out of STAMJ8R1 STAMJ8R2, STAMJ8R3, STAMJ8R4 and STAMJ8R5